Mathematical Stories

Mathematical Stories

Lecture Notes

July 15, 2026

Contents

Introduction

Purpose and Philosophy

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognises the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, formulate precise questions, distinguish an object from its notation, notice structure, and build simple but meaningful chains of reasoning. A solved problem is valuable not only because it produces an answer, but also because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorise, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as an answer to a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some habits of thought will deliberately return in different places. The same question of what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. A mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognise patterns of reasoning rather than only patterns of exercises.

The number of answers produced is not, by itself, a measure of progress. What matters is the quality of the work. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each important step was valid? Did we interpret or verify the result? These questions are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as the art of asking good questions and giving well-reasoned answers.

How the Course Works

This book is part of a weekly learning process rather than a collection of material to be read only before an examination. Each of the twelve thematic chapters is designed to support one week of study. A chapter provides the mathematical structure for the lecture, a complete narrative to which the student can return, and a starting point for individual problem solving and classroom discussion.

In a fifteen-week semester, the twelve thematic weeks establish the main rhythm of the course. The remaining time can be used to introduce the method of work, connect topics, revise important ideas, and present more substantial solutions.

The Weekly Learning Cycle

A student should not leave a topic behind when the lecture ends. A typical week consists of the following connected stages:

1. **Attend the lecture.** The lecture introduces the topic, motivates the central questions, and demonstrates how the relevant mathematical objects and methods should be read.

2. **Return to the chapter.** Read the corresponding chapter carefully after the lecture. The PDF is not merely a summary of slides. It is an organised narrative containing definitions, explanations, examples, interpretations, and mathematical arguments that can be examined more than once.

3. **Ask questions and clarify the material.** Mark every place where a definition, transition, calculation, or conclusion is not yet clear. Use the text, your own examples, classroom discussion, and appropriate digital tools to resolve these questions.

4. **Solve the complete assigned problem set.** Move from following an explanation to constructing arguments independently. Work on every assigned problem and prepare a separate, self-contained note for each one.

5. **Prepare professional mathematical notes.** Use precise notation, explanatory sentences, justified transitions, and a clear conclusion. The note should show not only the answer, but also how it follows from the given information, definitions, and earlier results.

6. **Present and discuss your work.** During exercise classes, be prepared to explain a solution, answer questions about its steps, and describe what you learned while preparing it.

These stages form one continuous process. Each stage prepares the next, and the presentation and feedback lead back to further reading, revision, and improved work.

Exercise Classes: Presentation, Feedback, and Revision

Presenting mathematics is part of learning mathematics. It reveals whether an argument has genuinely been understood and whether the student can communicate it clearly. During a presentation, the student should be able to read the notation, explain the role of each important formula, justify the method, and respond to questions without relying on AI to speak on the student's behalf.

Feedback given during exercise classes is intended for the whole group, not only for the student whose work is currently being presented. Everyone should follow the argument and listen carefully to the discussion. By comparing different examples, students learn what a complete solution looks like, which explanations make an argument clear, which steps require justification, and which weaknesses require revision.

Students should not focus only on comments addressed directly to their own work. Listening to feedback on the work of others provides a broader set of examples and develops an operational understanding of the course standards: what makes mathematical work clear, justified, and complete, and what still needs improvement.

The feedback concerns the mathematical work and its results, not the personal worth of the student. What is examined is the quality of the note, the understanding demonstrated in the argument, and the clarity of the presentation. Feedback should identify what already works and what should be changed in order to produce a stronger solution.

The first submitted version does not always have to be the final version. Students should learn to receive comments, reconsider their choices, correct errors, add missing explanations, and improve both the written note and its presentation. Revision is not evidence of failure. It is a normal part of serious intellectual work.

This process is important throughout university study and later in professional life. Students and professionals are given tasks, asked to solve problems, expected to present results, and required to respond to feedback and critical review. Learning to improve an initial solution is therefore not an additional classroom exercise. It is a fundamental habit of responsible academic and professional work.

Progress and Assessment

Progress is demonstrated by the quality and increasing independence of the student's work. By the end of a weekly cycle, the student should be able to identify the main mathematical objects, state the essential definitions, solve representative problems, justify the principal steps, verify or interpret the result, and communicate the argument clearly.

Assessment in the exercise component focuses primarily on how the student works throughout the semester. Evidence of serious engagement includes:

- regular preparation and completion of the assigned problem sets,
- documented attempts to solve problems rather than bare answers,
- active participation in presentations and classroom discussion,
- separate, self-contained notes for individual problems,
- careful revision of work after feedback,
- increasing clarity, independence, and mathematical precision.

A student is not defined by the quality of the first solution or presentation. Some students may initially find mathematical notation, formal writing, or public explanation difficult. A weak beginning does not prevent a very good final grade for the exercise component. A student who works conscientiously, responds to feedback, corrects earlier weaknesses, and demonstrates substantial development may finish the exercise component with a very good grade.

This approach does not lower the standard of mathematical work. It recognises that reaching the standard through sustained effort and revision is itself a meaningful achievement. The exercise-component grade reflects a process of learning, not merely a snapshot taken at the beginning of the semester.

Systematic work according to these guidelines provides the basis for earning a passing grade for the exercise component. The examination is assessed separately and tests the student's mathematical knowledge and ability to use it independently. Regular work does not replace the examination, but it develops the understanding and skills that substantially increase the likelihood of successfully completing the course.

How to Prepare Mathematical Notes

A mathematical note is not merely a place where an answer is recorded. Its purpose is to show how a problem is understood, how relevant information is selected, how a method is chosen, and how a conclusion follows from definitions, assumptions, and earlier results.

Mathematical expressions are compact. A single formula may contain several ideas at once: objects, operations, conditions, and logical relationships. Reading a formula therefore means more than pronouncing its symbols. A student should be able to explain what the objects represent, what operation is being performed, why the expression is meaningful, and what information the formula provides.

The same principle applies to writing. A sequence of formulas without explanatory sentences rarely communicates a complete mathematical argument. Formulas should be connected by words that explain their role. A good solution tells the reader why a calculation is being performed, which definition or theorem is being used, whether its assumptions are satisfied, and how the result answers the original question.

Make the Note Self-Contained

Prepare one separate, self-contained note for each assigned problem, including all of its subparts. Begin by writing the problem or a complete and faithful formulation of it. The reader should not need to search through another document to discover what is being solved. After several weeks, you should still be able to open the note and understand both the question and the solution.

Identify what is given and what must be found, proved, or explained. Introduce the relevant notation and determine what kinds of mathematical objects appear in the problem. Before calculating, ask what information is available and what would count as a satisfactory answer.

Make the Reasoning Visible

Present the principal idea or method before, or while, carrying out the calculation. Important transitions should be explained with complete sentences. For example:

- Because the determinant is non-zero, the matrix is invertible.
- We may therefore multiply both sides by the inverse matrix.
- By the definition of the derivative, we consider the following limit.
- Since the direction vector is non-zero, it determines a line.
- Substituting the result into the original equation verifies the solution.

Such sentences are not decoration around the mathematics. They are part of the mathematics because they show the logical relationship between successive formulas.

It is not necessary to describe every elementary arithmetic operation. The amount of explanation should be proportional to the problem. A simple exercise may require only a few sentences, while a more substantial problem may require a longer argument. The goal is not to make every note long. The goal is to make every essential step understandable and justified.

Write as briefly as possible, but as completely as necessary.

A Complete Solution

A complete note should make the following five stages visible:

1. **Understand the problem.** Record the problem, identify the given information, and state clearly what must be found, proved, or explained.
2. **Choose a method.** Select the definitions, theorems, or techniques that connect the available information with the goal.
3. **Construct the argument.** Carry out the necessary steps and connect formulas with sentences that explain why each important transition is valid.

4. **Verify the result.** Check calculations, assumptions, domain restrictions, or geometric meaning whenever appropriate.

5. **State the conclusion.** Answer the original question clearly and in its proper context.

A result without its context is not a complete solution. A bare final answer, an unexplained sequence of formulas, or a few unconnected calculations will be treated as incomplete work and should be revised.

Quality cannot be replaced by quantity, but quality does not remove the obligation to complete the assigned work. Students are expected to prepare separate, complete solutions for the entire assigned problem set. Producing many unexplained answers is insufficient, but preparing one polished solution while ignoring the remaining problems is also insufficient.

Working with AI

We are studying in an age in which AI tools are widely available. The free versions of these tools are sufficient for the kinds of support expected in this course: asking questions, clarifying notation, generating simpler examples, checking reasoning, identifying missing steps, and improving the organisation of mathematical writing. Access to expensive software is not required.

AI should be used as an individual learning assistant. It may help transform rough calculations into clear mathematical prose, but it does not replace the student's mathematical work. Receiving a well-written answer is not the same as understanding it. The student remains responsible for checking every definition, calculation, assumption, conclusion, and formula included in the submitted note.

Use AI with a focused context. Work on one problem at a time and prepare one complete note for that problem. Do not paste an entire problem set into a single conversation and request all answers at once. Handling many unrelated tasks together encourages brief, generic, and mathematically weak notes. Focused work makes it easier to question each step, identify missing justifications, and produce a solution that can be understood and presented.

The one-problem-at-a-time principle does not mean that only one problem from the set should be prepared. Students are expected to repeat this focused process for every problem in the complete assigned set.

A polished AI-generated note is insufficient if the student cannot read its formulas, explain its terminology, justify its transitions, or present the argument independently. When using AI, ask not only for a solution, but also questions that deepen understanding:

- What does this formula mean?
- What mathematical object does each symbol represent?
- Why is this step valid?
- Which definition or theorem is being used?
- Are the assumptions of the theorem satisfied?
- Is there a simpler way to explain this transition?
- How can the result be checked?
- What should I be able to explain during class?

Preparing notes in this way is part of learning mathematics. It develops the ability to read notation, recognise mathematical objects, organise information, construct arguments, and communicate ideas precisely. Repeating this process for the complete problem set builds the habits of systematic and conscientious work that should accompany the student throughout university study and later professional life.

Student GitHub Workspace

The current instructions, problem templates, and technical workflow for student work are maintained in the public repository `dchorazkiewicz/Math_Problems_Repo`. Students should create a public fork of this repository and commit their notes to their own fork. The course repository is the common template; it is not the place where individual student solutions should be submitted.

The repository contains a short map of the working process:

- `README.md` explains the purpose and structure of the workspace and provides the weekly index;
- `SOURCE_MATERIAL.md` links the current lecture PDF and maps each week to the corresponding LaTeX problem source used by an AI assistant;
- `NOTE_GUIDELINES.md` defines the standard of a complete mathematical note;
- `AI_WORKFLOW.md` describes the one-problem-at-a-time workflow with AI, review, commits, presentation, feedback, and revision;
- `AGENTS.md` gives a connected AI assistant the operational rules for working safely in the student's fork;
- the `docs/` directory contains twelve weekly folders and one Markdown file for each problem.

The workspace uses Markdown for student notes, Git and GitHub for version history and public forks, MathJax for mathematical typesetting, MkDocs for the web presentation, and GitHub Actions for automatic validation and deployment. These instructions and templates may be updated independently of this PDF, so students should consult the repository for the current working procedure. The repository links back to this PDF and to its LaTeX sources. Students use the PDF as the readable course material containing the theory, definitions, examples, and problem sets. A connected AI assistant uses the current mapped LaTeX file as the exact machine-readable source of a selected problem statement.

Chapter 1

Euclidean Geometry and Cartesian Coordinates

1.1 The Euclidean Geometry Available Before Coordinates

Before introducing coordinates, we recall the geometric world that is already available. At this stage there are no ordered pairs, no axes, and no vectors. There are points, straight lines, segments, circles, and angles, together with relations such as incidence, betweenness, congruence, perpendicularity, and parallelism.

Remark

Coordinates will later translate this geometry into arithmetic. They do not create the geometry. We can already construct and compare geometric objects before assigning numbers to all points and angles.

1.1.1 Three Different Levels Must Not Be Confused

We shall distinguish three kinds of statements.

1. *Euclid's postulates* describe the geometric world being assumed.
2. *Straightedge-and-compass rules* describe which elementary drawings and intersections may be used in a construction.
3. *Derived constructions* must be built from those elementary operations and justified by a proof.

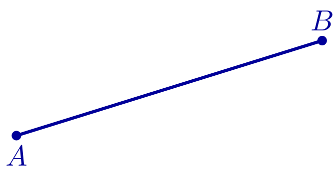
In particular, a perpendicular bisector, a perpendicular through a point, an angle bisector, or a parallel through a point will not be treated as primitive construction postulates. They will be constructed.

1.1.2 Euclid's Five Postulates

The following are the five postulates used in the first book of Euclid's *Elements*. The diagrams illustrate their meaning; they are not proofs.

Postulate I: Joining Two Points

A straight segment can be drawn from any point to any other point.



a segment is drawn from A to B

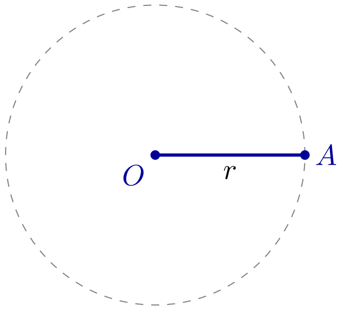
Postulate II: Extending a Segment

A finite straight segment can be extended continuously in the same straight line.

AB is extended beyond B

Postulate III: Drawing a Circle

A circle can be drawn with any chosen centre and any chosen radius.

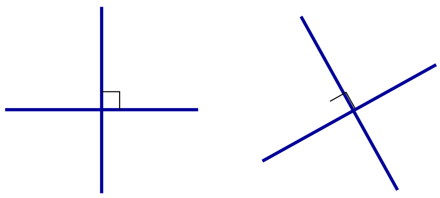


centre O and radius r determine a circle

This postulate is what permits a length to be copied geometrically without first assigning a number to it.

Postulate IV: Equality of Right Angles

All right angles are congruent.

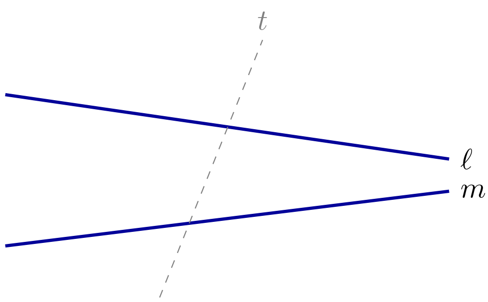


every right angle represents the same geometric magnitude

This postulate does not say that a perpendicular may simply be drawn without construction. It says that once right angles have been constructed, they are all equal.

Postulate V: The Parallel Postulate

If a transversal meets two straight lines and the interior angles on one side sum to less than two right angles, then the two lines, when extended, meet on that side.



the angle condition determines on which side ℓ and m meet

A familiar modern equivalent is Playfair's formulation: through a point outside a given line there passes exactly one parallel to that line. We shall use that form later, after the relation between the two formulations has been made clear.

1.1.3 Elementary Rules for Straightedge-and-Compass Work

For explicit constructions we use the following operational rules.

1. Through two already constructed distinct points, draw the straight line.
2. With an already constructed point as centre and an already constructed segment as radius, draw a circle.
3. Retain intersection points of two constructed lines, a line and a circle, or two circles whenever such intersections exist.

The first two rules are the operational forms of Euclid's first three postulates; the third explains how new constructible points are obtained. These rules are the complete elementary toolkit used in the constructions below.

1.1.4 Constructions Derived from the Toolkit

The Perpendicular Bisector and the Midpoint

Let A and B be distinct points. Draw two circles with the same radius, one centred at A and one centred at B, choosing the radius larger than half of $|AB|$. Let their intersection points be P and Q.

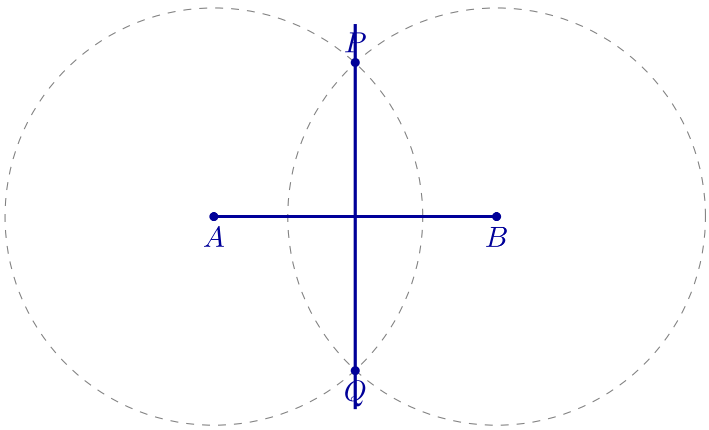
Because P lies on both circles,

$$|PA| = |PB|,$$

and similarly,

$$|QA| = |QB|.$$

Thus both P and Q are equidistant from A and B. The line PQ is the perpendicular bisector of AB.



The intersection of PQ with AB is the midpoint of AB. Thus midpoint and perpendicularity have been constructed from equal radii and intersections.

A Perpendicular Through a Point on a Line

Let P lie on a line ℓ.

Step 1. Draw a circle centred at P. It meets ℓ at two points A and B. Since A and B lie on the same circle,

$$|PA| = |PB|.$$

Thus P is the midpoint of AB.

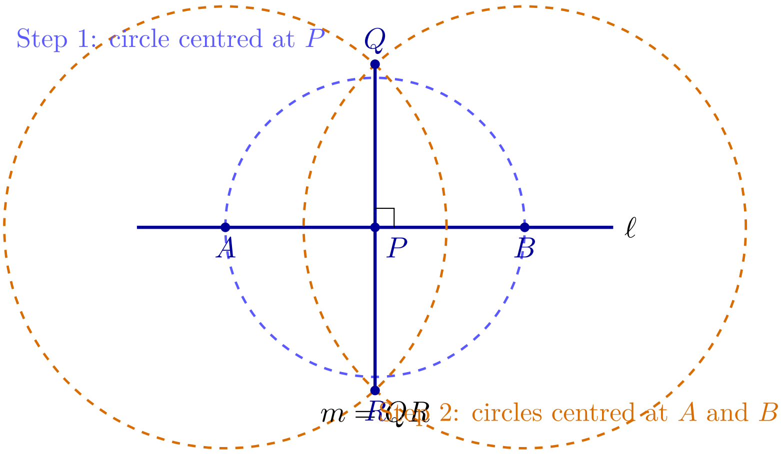
Step 2. Draw two circles with the same radius, one centred at A and one centred at B, choosing the radius larger than $|AP|$. Let their intersection points be Q and R. Then

$$|AQ| = |BQ| \qquad \text{and} \qquad |AR| = |BR|.$$

Both Q and R are therefore equidistant from A and B, so the line QR is the perpendicular bisector of AB. Because P is the midpoint of AB, the line QR passes through P. Since AB lies on ℓ,

$$QR \perp AB, \qquad \text{hence} \qquad QR \perp \ell.$$

The required perpendicular is the line $m = QR$.



The perpendicular has now been obtained entirely from the permitted straightedge-and-compass operations.

A Perpendicular Through a Point Outside a Line

Let P lie outside a line ℓ.

Step 1. Draw a circle centred at P with radius large enough to meet ℓ at two points A and B. Since A and B lie on the same circle,

$$|PA| = |PB|.$$

Step 2. Draw two circles of radius $|PA|$: one centred at A and one centred at B. The two circles meet at P and at a second point Q. Therefore

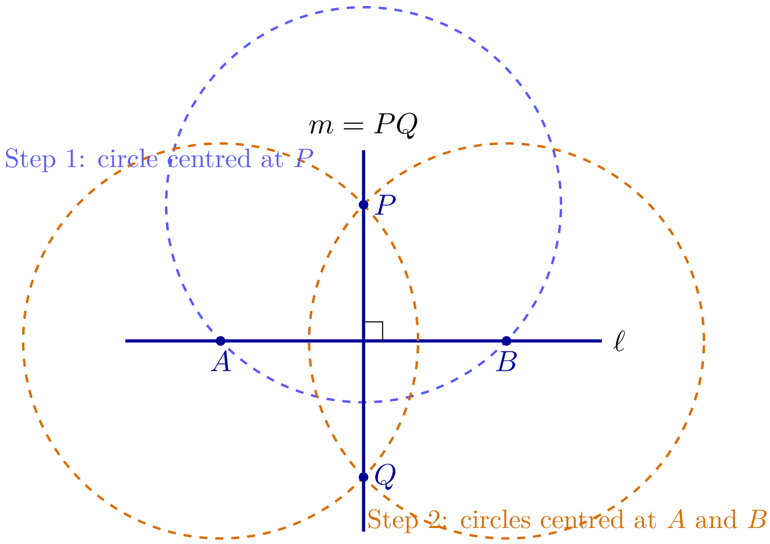
$$|AP| = |BP| \qquad \text{and} \qquad |AQ| = |BQ|.$$

Thus both P and Q are equidistant from A and B. The line PQ is therefore the perpendicular bisector of AB.

Because A and B lie on ℓ, the segment AB is contained in ℓ. Consequently,

$$PQ \perp AB, \qquad \text{hence} \qquad PQ \perp \ell.$$

The required perpendicular is the line $m = PQ$.



The perpendicular has now been obtained entirely from the permitted straightedge-and-compass operations: drawing circles, retaining their intersection points, and drawing the line through two constructed points.

Bisecting an Angle

Let two rays with common endpoint O form an angle. Draw a circle centred at O meeting the rays at A and B. Then draw equal-radius circles centred at A and B, and let one of their intersections inside the angle be P. Since

$$|OA| = |OB|, \qquad |AP| = |BP|, \qquad |OP| = |OP|,$$

the triangles OAP and OBP are congruent. Therefore

$$\angle AOP = \angle POB.$$

The ray OP is the angle bisector.

Constructing a Parallel Through a Point

Let P lie outside a line ℓ. First construct a line m through P perpendicular to ℓ. Then construct a line n through P perpendicular to m.

The transversal m forms equal corresponding right angles with n and ℓ. By the converse of the corresponding-angles theorem, the two lines are parallel:

$$n \parallel \ell.$$

The fifth postulate, equivalently Playfair's postulate, guarantees that this is the unique parallel to ℓ through P.

1.1.5 Parallel Lines and Proportional Segments

A central fact of Euclidean geometry is that parallel lines preserve ratios. This is the content of Thales' theorem in the form needed here.

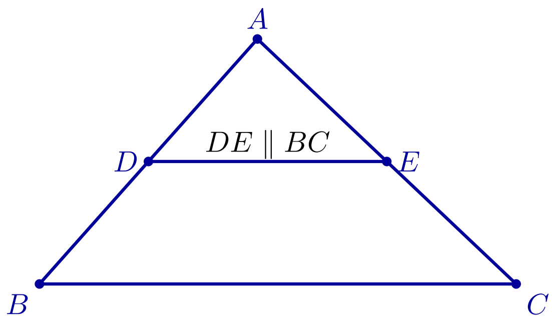
Theorem

Let D lie on AB and E lie on AC in a triangle ABC. If

$$DE \parallel BC,$$

then

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}.$$



The smaller triangle ADE and the larger triangle ABC have the same three angles. Their corresponding sides differ by one common scale factor. This is the basic phenomenon of similarity.

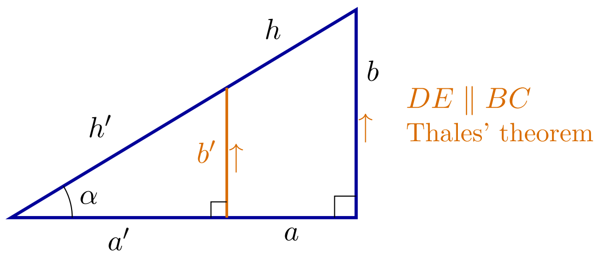
Definition

Two triangles are similar when their corresponding angles are equal. In that case their corresponding side lengths are proportional.

Similarity lets us change the size of a figure while preserving its shape. It is the reason ratios of corresponding lengths can depend only on angles.

1.1.6 Sine and Cosine Before Coordinates

Take a right triangle containing an acute angle α. Let h denote the hypotenuse, let a be the leg adjacent to α, and let b be the leg opposite α.



$$\frac{a'}{a} = \frac{b'}{b} = \frac{h'}{h}, \qquad \frac{a'}{h'} = \frac{a}{h}, \qquad \frac{b'}{h'} = \frac{b}{h}.$$

For an acute geometric angle α, define

$$\cos\alpha = \frac{a}{h}, \qquad \sin\alpha = \frac{b}{h}.$$

These ratios are independent of the chosen right triangle because all right triangles containing the same acute angle are similar.

For angles occurring in an arbitrary Euclidean triangle, we also need the right and obtuse cases. We set

$$\cos(\text{right angle}) = 0.$$

If α is obtuse and β is its acute supplement, so that together they form a straight angle, define

$$\cos \alpha = -\cos \beta.$$

Thus the geometric cosine is defined for every angle between 0 and a straight angle. For acute angles it is positive, for a right angle it is zero, and for obtuse angles it is negative.

The essential role of Thales' theorem and similarity is that these values depend only on the angle, not on the size of the triangle used to represent it.

For an acute angle, the Pythagorean theorem gives

$$a^2 + b^2 = h^2.$$

Dividing by h^2 yields

$$\left(\frac{a}{h}\right)^2 + \left(\frac{b}{h}\right)^2 = 1,$$

and hence

$$\cos^2 \alpha + \sin^2 \alpha = 1.$$

Thus the fundamental trigonometric identity is a direct reformulation of the Pythagorean theorem.

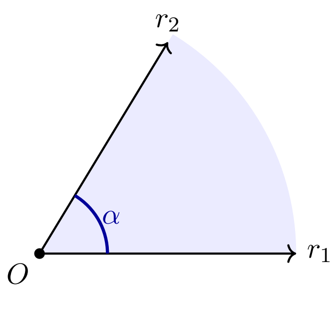
1.1.7 Angles and Their Numerical Measure

An angle exists geometrically before it is assigned a number. We can compare, copy, add, and bisect angles without degrees or radians. A numerical angle measure requires an additional choice of unit.

Definition

Let r_1 and r_2 be two rays with the same initial point O. The rays, together with a choice of one of the two regions between them, determine an angle α with vertex O.

The angle is the chosen geometric region. It is neither the small arc drawn near the vertex nor a number. The arc is only a graphical mark indicating which region has been selected.



Why circles give a consistent measure

Draw a circle centred at O. The rays cut from the circle an arc corresponding to the chosen angle.

Theorem

For every Euclidean circle, the ratio of circumference C to diameter D is the same constant:

$$\frac{C}{D} = \pi.$$

This constant is called π. Since $D = 2r$, every circle of radius r has circumference

$$C = 2\pi r.$$

The constancy of C/D follows from similarity: enlarging a circle by a factor λ multiplies every length, including its diameter, circumference, and corresponding arc lengths, by λ.

If the same two rays meet two circles of radii r_1 and r_2 and cut arc lengths s_1 and s_2, then

$$\frac{s_1}{s_2} = \frac{r_1}{r_2},$$

and therefore

$$\frac{s_1}{r_1} = \frac{s_2}{r_2}.$$

Thus the quotient s/r depends only on the angle, not on the chosen circle.

Definition

Let s be the length of the selected arc on a circle of radius r centred at the vertex. The radian measure of α is

$$m_{\text{rad}}(\alpha) = \frac{s}{r}.$$

For a full turn, $s = 2\pi r$, so

$$m_{\text{rad}}(\text{full turn}) = 2\pi.$$

A half-turn has radian measure π, and a quarter-turn has radian measure $\pi/2$.

Degrees are a second numerical scale. A full turn is assigned 360°, so

$$2\pi \text{ radians} = 360^\circ$$

and

$$\boxed{m_{\text{deg}}(\alpha) = \frac{180^\circ}{\pi}\, m_{\text{rad}}(\alpha)}.$$

In particular, a constructible right angle has the numerical measures

$$m_{\text{rad}}(\text{right angle}) = \frac{\pi}{2}, \qquad m_{\text{deg}}(\text{right angle}) = 90^\circ.$$

The angle is still the geometric region; $\pi/2$ and 90° are two numbers assigned to it by two different measurement systems.

Sine and cosine as functions of a number

The trigonometric ratios were first defined for acute geometric angles. We now extend them to a full turn. For $t \in [0, 2\pi)$, let $\beta \in [0, \pi/2]$ be the acute reference angle between the direction of t and the nearest horizontal ray. The magnitudes $\sin\beta$ and $\cos\beta$ come from the right-triangle definition; their signs are determined by the quadrant:

	$\cos t$	$\sin t$
$0 \le t \le \pi/2$	$+\cos\beta$	$+\sin\beta$
$\pi/2 \le t \le \pi$	$-\cos\beta$	$+\sin\beta$
$\pi \le t \le 3\pi/2$	$-\cos\beta$	$-\sin\beta$
$3\pi/2 \le t < 2\pi$	$+\cos\beta$	$-\sin\beta$

At the axes this gives the familiar values 0, 1, and -1. For arbitrary real t, define sine and cosine periodically:

$$\sin(t + 2\pi k) = \sin t, \qquad \cos(t + 2\pi k) = \cos t, \qquad k \in \mathbb{Z}.$$

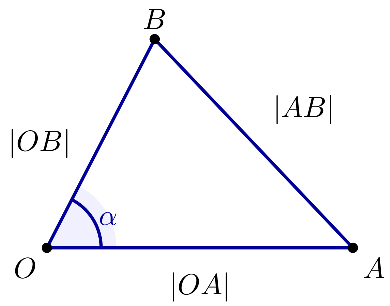
If $t = m_{\text{rad}}(\alpha)$, then $\cos t$ is the same value as the geometric cosine $\cos \alpha$. This explains the notation $\cos(m_{\text{rad}}(\alpha))$.

The Euclidean law of cosines

Directly measuring an arc is often difficult. A triangle provides another way to recover the measure of an angle. Choose points

$$A \in r_1 \setminus \{O\}, \qquad B \in r_2 \setminus \{O\}.$$

The points form the Euclidean triangle OAB.



adjacent to α:
$|OA|$ and $|OB|$

opposite α:
$|AB|$

Theorem

If $\alpha = \angle AOB$, then the side lengths of the Euclidean triangle satisfy

$$|AB|^2 = |OA|^2 + |OB|^2 - 2|OA|\,|OB| \cos \alpha.$$

Equivalently, the last factor may be written as $\cos(m_{\text{rad}}(\alpha))$.

The theorem belongs to Euclidean geometry; it does not require Cartesian coordinates. Solving it for the angle measure gives

$$\boxed{m_{\text{rad}}(\alpha) = \arccos\left(\frac{|OA|^2 + |OB|^2 - |AB|^2}{2|OA|\,|OB|}\right)}.$$

Later, coordinates will provide a convenient arithmetic method for calculating the three Euclidean lengths $|OA|$, $|OB|$, and $|AB|$.

Remark

At this point no vector has been introduced. We have only Euclidean objects, postulates, constructions, similarity, trigonometric ratios, numerical angle measure, and the Euclidean law of cosines. Vectors will appear only in the next chapter, after the Cartesian coordinate system has been constructed.

The next step is to choose two perpendicular lines, an origin, directions, and a unit length. Those choices will convert the Euclidean plane into a Cartesian coordinate plane.

1.2 Constructing Cartesian Coordinates from Euclidean Geometry

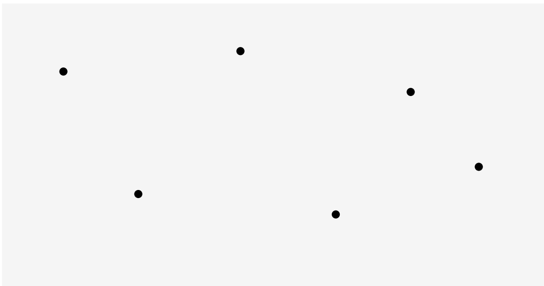
We begin with a Euclidean plane, not with ordered pairs and not with a coordinate system that has already been drawn. Points, straight lines, right angles, perpendicularity, congruent segments, and the Pythagorean theorem belong to the geometry available before coordinates are introduced. Our task is to construct, step by step, the additional choices that turn this plane into a Cartesian coordinate plane.

Remark

Coordinates do not create Euclidean geometry. They arise after we choose and construct a particular system for assigning numerical labels to points of an already existing Euclidean plane.

1.2.1 The Plane Before Coordinates

At first there is only a Euclidean plane, with no distinguished line, origin, directions, or numerical scale.



a Euclidean plane with no chosen coordinates

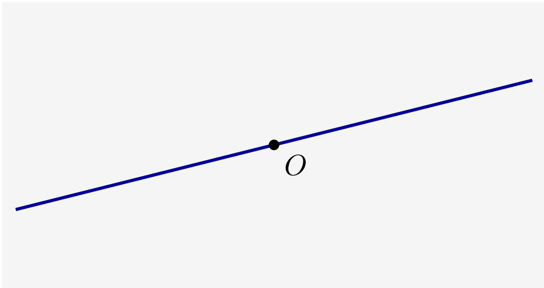
1.2.2 First Construction: A Straight Line

Choose two distinct points of the plane and construct the unique straight line through them. At this stage it is only a chosen Euclidean line. It is not yet an axis, because there is not yet a second line, an origin, a direction, or a scale.

the first constructed line

1.2.3 Second Construction: A Point on the Line

Choose an arbitrary point O on the constructed line. This point will later become the common zero of both coordinate axes, but it becomes an origin only after the rest of the coordinate structure has been constructed.

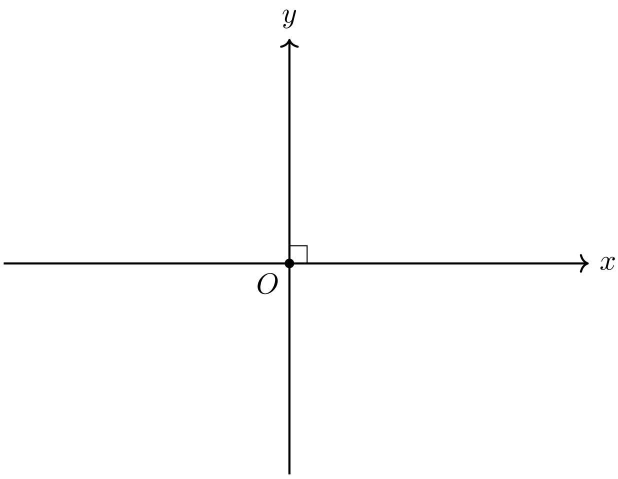


a point chosen on the first line

1.2.4 Third Construction: The Perpendicular Line

Through O, construct the unique line perpendicular to the first line. The two constructed lines now intersect at a right angle. Only now do we have the geometric framework from which two coordinate axes can be made.

We call the first line the x-axis, the perpendicular line the y-axis, and their intersection O the origin.

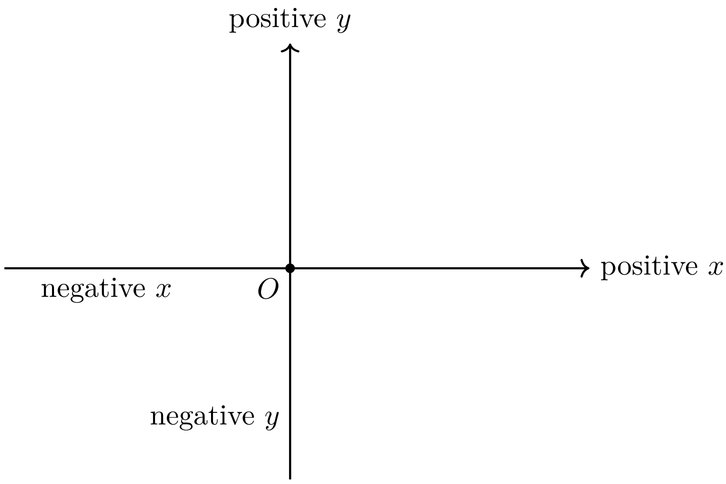


The axes are perpendicular because this was required in their geometric construction. No coordinate formula and no numerical angle measure has been used.

1.2.5 Choosing Positive and Negative Directions

Each axis is divided by O into two opposite rays. Choose one ray on the x-axis as positive and one ray on the y-axis as positive. The opposite rays are then called negative.

These choices are not forced by the Euclidean plane. Reversing one of them would produce a different coordinate system on the same geometric plane.

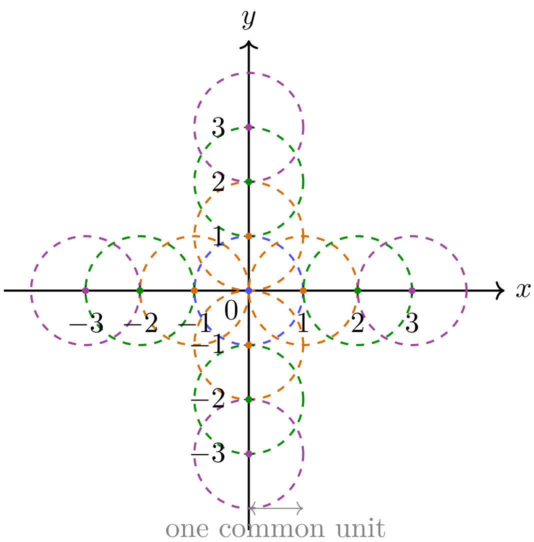


1.2.6 Constructing a Common Unit Length

Choose a positive segment length and call it one unit. Using a compass, transfer this same length from O repeatedly along both axes and in both directions. The first marks are labelled 1 and -1; repetition constructs the integer marks

$$\dots, -3, -2, -1, 0, 1, 2, 3, \dots$$

on each axis.



Remark

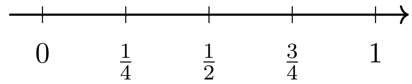
The same unit must be used on both axes. A segment labelled as one unit on the x-axis must be congruent to a segment labelled as one unit on the y-axis. This common scale is what later allows the Pythagorean theorem to produce the standard coordinate formula for Euclidean distance.

1.2.7 Refining the Scale by Bisection

The compass and straightedge construction does not stop at integer marks. Bisecting each unit segment constructs halves. Bisecting again constructs quarters, then eighths, and so on. Repeating this process on every integer interval and on both sides of the origin gives all dyadic marks

$$\frac{k}{2^n}, \qquad k \in \mathbb{Z}, \quad n \in \mathbb{N}.$$

successive bisections of one unit



The dyadic marks are dense on the axis: every interval, however small, contains such marks. They therefore provide arbitrarily fine numerical approximations to positions on the line.

Definition

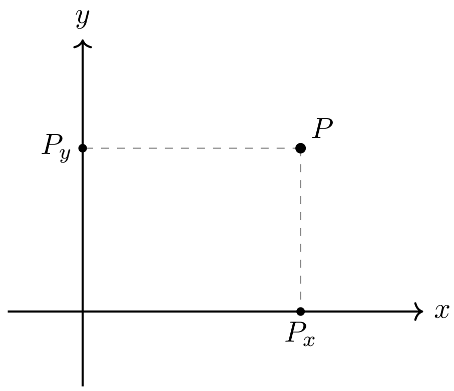
Continuity assumption. Once an origin, a positive direction, and a unit length are fixed on a Euclidean line, every point of the line corresponds to exactly one real number, and every real number corresponds to exactly one point of the line.

This assumption is additional to the finite straightedge-and-compass operations. Those operations construct many particular positions, including all dyadic marks, but not every real position. Continuity supplies the complete real scale used by coordinates. After this step, every point of each axis has a unique signed real label.

1.2.8 Assigning Coordinates by Orthogonal Projection

Only after the axes, directions, common unit, integer marks, refined scale, and real labels have been constructed can an arbitrary point of the plane be given coordinates.

Let P be any point of the plane. Through P, construct the line perpendicular to the x-axis. It meets the x-axis at a unique point P_x. Through P, construct the line perpendicular to the y-axis. It meets the y-axis at a unique point P_y.



If the signed real labels of P_x and P_y are x and y, respectively, we assign to P the ordered pair

$$P \longleftrightarrow (x, y).$$

The construction is unambiguous because the perpendicular through a given point to a given line is unique. Conversely, given real numbers x and y, locate the corresponding points on the two axes and construct through them lines perpendicular to their respective axes. Those two lines meet in exactly one point of the plane.

Theorem

Once the two perpendicular axes, their positive directions, and a common unit are fixed, orthogonal projection gives a one-to-one correspondence

$$\text{points of the Euclidean plane} \longleftrightarrow \text{pairs } (x, y) \in \mathbb{R}^2.$$

The ordered pair is the numerical address of a geometric point in the chosen coordinate system. It is not the point itself and it is not the source of the geometry.

1.2.9 The Coordinate Formula for Euclidean Distance

The Euclidean distance between two points is the length of the straight segment joining them. Coordinates now allow this geometric length to be calculated.

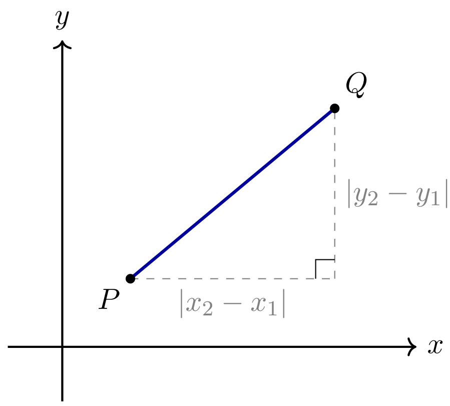
For points

$$P = (x_1, y_1), \qquad Q = (x_2, y_2),$$

their projections show that the horizontal and vertical separations have lengths

$$|x_2 - x_1| \qquad \text{and} \qquad |y_2 - y_1|.$$

Together with the segment PQ, these two perpendicular segments form a right triangle. Here "right" refers to the Euclidean right angle already used in the construction; no numerical measure of that angle is required.



By the Pythagorean theorem,

$$d_E(P, Q)^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2,$$

and therefore

$$\boxed{d_E(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}.$$

The formula does not create a new notion of distance. It is the numerical coordinate expression of the Euclidean segment length already present in the plane.

Summary

The construction proceeds in the following order:

1. begin with a Euclidean plane containing no chosen coordinates;
2. construct one straight line;
3. choose a point O on that line;
4. construct through O the perpendicular line, and only then name the two lines as coordinate axes and O as their origin;
5. choose positive directions on both axes;
6. choose one common unit length and transfer it with a compass in both positive and negative directions to construct the integer marks;

7. repeatedly bisect intervals to construct the dense family of dyadic marks;
8. use Euclidean continuity to identify every axis position with a real number;
9. assign to every point its unique pair of orthogonal projections;
10. derive the coordinate distance formula from the resulting right triangle and the Pythagorean theorem.

Chapter 2

Vectors, Bases, and Coordinate Systems

2.1 Vectors from Coordinate Displacements

A coordinate pair tells us where a point is. It does not yet tell us how one point is related to another. To describe motion from one point to another we need a new object: not another position, but a record of the change of position.

Choose

$$A = (1, 2), \qquad B = (4, 6).$$

To travel from A to B, the horizontal coordinate must increase by 3 and the vertical coordinate by 4. The pair

$$(3, 4)$$

therefore records an instruction:

move three units in the positive x direction and four units in the positive y direction.

This instruction can be carried out from A, but it can also be carried out from any other point. The starting point may change while the displacement remains the same. That distinction is the idea from which vectors arise.

The development below follows a fixed question-driven order. We first decide when two arrows represent the same displacement. We then ask how successive displacements combine, what it means to repeat or reverse one, what addition and scaling can build, how to measure length and direction, and how two displacements can be compared geometrically. The coordinate formulas will be consequences of those questions, not isolated rules introduced for memorisation.

2.1.1 A Vector Is a Displacement, Not a Place

Let

$$A = (x_1, y_1), \qquad B = (x_2, y_2).$$

The point A is a position and the point B is another position. The arrow from A to B asks a different question:

what change of coordinates carries the starting point to the endpoint?

The horizontal change is $x_2 - x_1$ and the vertical change is $y_2 - y_1$. Therefore

$$\overrightarrow{AB} = (x_2 - x_1,\, y_2 - y_1).$$

This subtraction is not an arbitrary convention. It is forced by the equations

$$x_1 + \Delta x = x_2, \qquad y_1 + \Delta y = y_2.$$

Solving them gives

$$\Delta x = x_2 - x_1, \qquad \Delta y = y_2 - y_1.$$

Definition

A vector in the Cartesian plane is a coordinate displacement

$$\mathbf{v} = (v_1, v_2).$$

It instructs us to change the first coordinate by v_1 and the second coordinate by v_2.

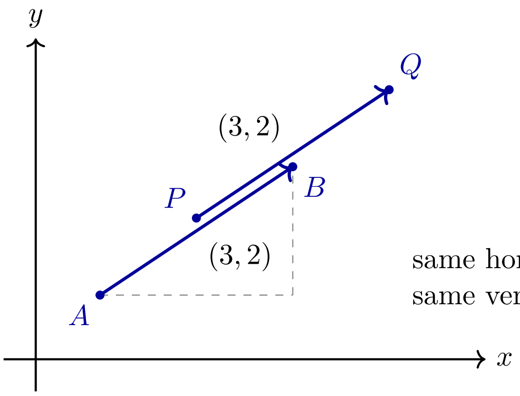
A vector is therefore not tied to one starting point. If $P = (p_1, p_2)$ and we apply the same displacement $\mathbf{v} = (v_1, v_2)$, then the endpoint must be

$$Q = (p_1 + v_1,\, p_2 + v_2),$$

and indeed

$$\overrightarrow{PQ} = (v_1, v_2) = \mathbf{v}.$$

Two arrows drawn in different places represent the same vector when they have the same coordinate changes. Their positions may differ; their displacement is the same.



same horizontal change
same vertical change

The zero vector

$$\mathbf{0} = (0, 0)$$

means no change at all. Starting anywhere and applying $\mathbf{0}$ leaves the point fixed.

The opposite vector

$$-\mathbf{v} = (-v_1, -v_2)$$

undoes the original displacement. If $\mathbf{v}$ carries A to B, then $-\mathbf{v}$ carries B back to A. Thus

$$\overrightarrow{BA} = -\overrightarrow{AB}.$$

2.1.2 Two Journeys Become One: Why Vectors Add

Suppose a point is moved twice. The first displacement is

$$\mathbf{u} = (u_1, u_2),$$

and the second is

$$\mathbf{v} = (v_1, v_2).$$

We want one vector that has exactly the same final effect as these two moves performed successively.

Start at an arbitrary point (x, y). After applying $\mathbf{u}$ we are at

$$(x + u_1,\, y + u_2).$$

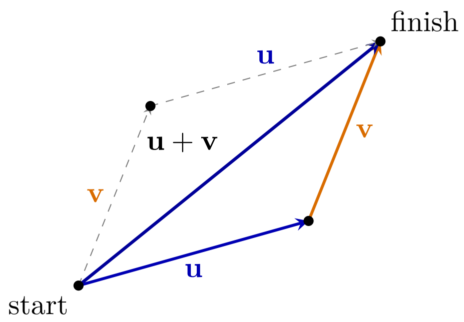
Applying $\mathbf{v}$ from there gives

$$(x + u_1 + v_1,\, y + u_2 + v_2).$$

Relative to the original point, the total horizontal change is $u_1 + v_1$ and the total vertical change is $u_2 + v_2$. Therefore the single displacement with the same effect is forced to be

$$\boxed{\mathbf{u} + \mathbf{v} = (u_1 + v_1,\, u_2 + v_2)}.$$

The formula is simply bookkeeping for a journey performed in two stages. It adds changes of the same kind: horizontal change to horizontal change and vertical change to vertical change.



The diagram contains two routes to the same endpoint:

$$\mathbf{u} \text{ then } \mathbf{v}, \qquad \mathbf{v} \text{ then } \mathbf{u}.$$

They agree because addition of real numbers is commutative:

$$u_1 + v_1 = v_1 + u_1, \qquad u_2 + v_2 = v_2 + u_2.$$

Hence

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}.$$

Geometrically this is the parallelogram rule. Algebraically it is the fact that the order of adding coordinate changes does not affect the final position.

Subtraction answers a different practical question: what displacement must be performed after $\mathbf{v}$ in order to have the same effect as $\mathbf{u}$? We seek a vector $\mathbf{w}$ satisfying

$$\mathbf{v} + \mathbf{w} = \mathbf{u}.$$

Solving coordinate by coordinate gives

$$\boxed{\mathbf{u} - \mathbf{v} = (u_1 - v_1,\, u_2 - v_2)}.$$

In particular, the displacement from A to B is the position of B minus the position of A:

$$\overrightarrow{AB} = B - A.$$

The expression does not subtract geometric points as physical objects. It subtracts their coordinate addresses to recover the change carrying one to the other.

2.1.3 Stretching, Shrinking, and Reversing a Displacement

Addition combines different displacements. A second operation appears when the same displacement is repeated.

If

$$\mathbf{v} = (v_1, v_2),$$

then performing it twice gives

$$\mathbf{v} + \mathbf{v} = (2v_1, 2v_2).$$

Performing it three times gives

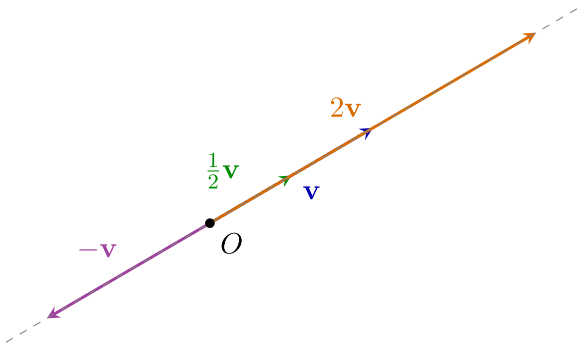
$$3\mathbf{v} = (3v_1, 3v_2).$$

The coordinates are multiplied by the same number because both components must be stretched by the same factor if the direction is to remain unchanged.

This suggests the general definition

$$\boxed{\lambda\mathbf{v} = (\lambda v_1, \lambda v_2), \qquad \lambda \in \mathbb{R}}.$$

The number λ is called a scalar. It does not provide a new direction; it changes how much of the old displacement is taken.



The picture separates the possible cases:

- if $\lambda > 1$, the vector is stretched;
- if $0 < \lambda < 1$, it is shortened;
- if $\lambda = 0$, it collapses to the zero vector;
- if $\lambda < 0$, it is also reversed.

In every case the result lies on the same line through the origin as $\mathbf{v}$. This gives a first geometric meaning of scalar multiples: nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel precisely when one is a scalar multiple of the other.

Scalar multiplication also explains why fractions and real factors are natural. The vector $\frac{1}{2}\mathbf{v}$ is the displacement that must be repeated twice to produce $\mathbf{v}$, while $\sqrt{2}\,\mathbf{v}$ has the same direction and $\sqrt{2}$ times the length. The operation extends the simple idea of repetition to continuous scaling.

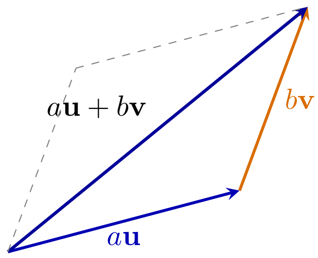
2.1.4 Linear Combinations: What Addition and Scaling Can Build

Addition and scalar multiplication were introduced as two separate operations, but they are most useful when they are combined. Given vectors $\mathbf{u}$ and $\mathbf{v}$ and real numbers a and b, the vector

$$a\mathbf{u} + b\mathbf{v}$$

is called a *linear combination* of $\mathbf{u}$ and $\mathbf{v}$.

The phrase has a direct geometric meaning. First scale $\mathbf{u}$ by a and $\mathbf{v}$ by b; then add the two resulting displacements. Thus a linear combination is not a new operation. It is a controlled use of the two operations already understood.



A single nonzero vector can generate only one line through the origin:

$$\{\lambda\mathbf{v} : \lambda \in \mathbb{R}\}.$$

Changing λ changes the length and possibly the orientation, but never the underlying line of motion.

Two vectors behave differently. If $\mathbf{u}$ and $\mathbf{v}$ are parallel, then every linear combination still lies on one line. If they are not parallel, then their linear combinations fill the whole plane. Geometrically, the two independent directions allow us to reach any endpoint by a suitable parallelogram.

Theorem

Two nonparallel vectors $\mathbf{u}$ and $\mathbf{v}$ determine every vector in the plane uniquely in the form

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}.$$

The numbers a and b record how much motion occurs in each chosen direction.

This observation is the bridge to the later notion of a basis. A basis is not an unrelated new object: it is a pair of directions whose linear combinations can describe every vector uniquely.

2.1.5 How Much Motion and in Which Direction?

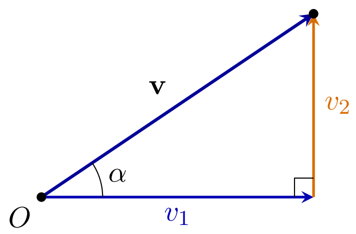
The coordinate pair of a vector records two changes, but it does not immediately answer two basic geometric questions:

1. How long is the displacement?
2. In which direction does it point?

Let

$$\mathbf{v} = (v_1, v_2)$$

be drawn from the origin. Its endpoint, together with its horizontal and vertical coordinate changes, forms a right triangle. The legs have lengths $|v_1|$ and $|v_2|$, while the vector itself is the hypotenuse.



By the Pythagorean theorem,

$$\|\mathbf{v}\|^2 = v_1^2 + v_2^2,$$

so the nonnegative length of the vector is

$$\boxed{\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}}.$$

The double bars distinguish the vector from the number measuring its length. In particular,

$$\|\mathbf{0}\| = 0,$$

and every nonzero vector has positive length.

This formula also reconnects vectors with the Euclidean distance between points. Since

$$\overrightarrow{AB} = B - A,$$

the length of the displacement from A to B is

$$\boxed{d_E(A, B) = \|B - A\|}.$$

A vector therefore packages both the direction from one point to another and the distance travelled.

Now assume $\mathbf{v} \neq \mathbf{0}$ and let α be its angle of inclination, measured from the positive x-axis. The right triangle gives

$$\cos\alpha = \frac{v_1}{\|\mathbf{v}\|}, \qquad \sin\alpha = \frac{v_2}{\|\mathbf{v}\|}.$$

Rearranging,

$$v_1 = \|\mathbf{v}\|\cos\alpha, \qquad v_2 = \|\mathbf{v}\|\sin\alpha.$$

Hence

$$\boxed{\mathbf{v} = \|\mathbf{v}\|(\cos\alpha, \sin\alpha)}.$$

This formula separates the two pieces of information carried by a nonzero vector:

- $\|\mathbf{v}\|$ says how much displacement occurs;
- $(\cos\alpha, \sin\alpha)$ is a unit vector saying in which direction.

Indeed,

$$\|(\cos\alpha, \sin\alpha)\| = \sqrt{\cos^2\alpha + \sin^2\alpha} = 1.$$

Thus every nonzero vector is a positive length multiplied by a unit direction. This decomposition will make comparison of two vectors by their angle possible in the next subsection.

2.1.6 One Angle Calculation, Two Geometric Tests

Take two arbitrary nonzero vectors

$$\mathbf{u} = (u_1, u_2), \qquad \mathbf{v} = (v_1, v_2).$$

Let their angles of inclination be α and β. Then

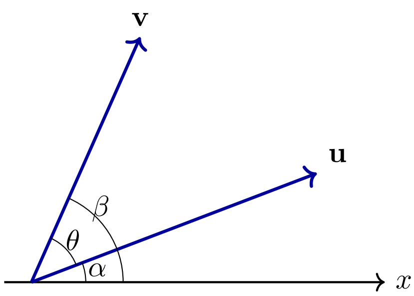
$$\mathbf{u} = \|\mathbf{u}\|(\cos\alpha, \sin\alpha),$$

$$\mathbf{v} = \|\mathbf{v}\|(\cos\beta, \sin\beta).$$

The difference $\beta - \alpha$ measures the directed turn from $\mathbf{u}$ to $\mathbf{v}$. The ordinary angle between the vectors will be denoted by

$$\theta = |\beta - \alpha|$$

when the smaller angle is intended.



Parallel directions arise from the sine of the angle difference

Two nonzero vectors determine the same unoriented direction precisely when the turn from one to the other is an integer multiple of a half-turn:

$$\beta - \alpha \equiv 0 \pmod{\pi}.$$

Equivalently,

$$\sin(\beta - \alpha) = 0.$$

Using the sine-difference identity,

$$\sin(\beta - \alpha) = \sin\beta\cos\alpha - \cos\beta\sin\alpha.$$

Now substitute

$$\cos\alpha = \frac{u_1}{\|\mathbf{u}\|}, \qquad \sin\alpha = \frac{u_2}{\|\mathbf{u}\|},$$
$$\cos\beta = \frac{v_1}{\|\mathbf{v}\|}, \qquad \sin\beta = \frac{v_2}{\|\mathbf{v}\|}.$$

We obtain

$$\begin{aligned}\sin(\beta - \alpha) &= \frac{v_2}{\|\mathbf{v}\|}\frac{u_1}{\|\mathbf{u}\|} - \frac{v_1}{\|\mathbf{v}\|}\frac{u_2}{\|\mathbf{u}\|} \\ &= \frac{u_1v_2 - u_2v_1}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.\end{aligned}$$

The denominator is nonzero. Therefore

$$\sin(\beta - \alpha) = 0 \iff u_1v_2 - u_2v_1 = 0.$$

We have discovered the coordinate test

$$\mathbf{u} \parallel \mathbf{v} \iff u_1v_2 - u_2v_1 = 0.$$

The same condition implies that one vector is a scalar multiple of the other. Indeed, if $v_1 \neq 0$, set $\lambda = u_1/v_1$. The equality

$$u_1 v_2 - u_2 v_1 = 0$$

gives $u_2 = \lambda v_2$, and hence

$$\mathbf{u} = \lambda \mathbf{v}.$$

If $v_1 = 0$, then $v_2 \neq 0$ and the same argument uses $\lambda = u_2/v_2$. Thus

$$\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} = \lambda \mathbf{v} \quad \text{for some } \lambda \neq 0.$$

This scalar-multiple criterion is a conclusion of the angle calculation, not a rule imposed in advance.

The cosine of the same difference reveals perpendicularity

The vectors are perpendicular precisely when their angle difference is a quarter-turn modulo a half-turn:

$$\beta - \alpha \equiv \frac{\pi}{2} \pmod{\pi}.$$

Equivalently,

$$\cos(\beta - \alpha) = 0.$$

The cosine-difference identity gives

$$\cos(\beta - \alpha) = \cos\beta\cos\alpha + \sin\beta\sin\alpha.$$

Substituting the coordinate ratios gives

$$\begin{aligned}\cos(\beta - \alpha) &= \frac{v_1}{\|\mathbf{v}\|}\frac{u_1}{\|\mathbf{u}\|} + \frac{v_2}{\|\mathbf{v}\|}\frac{u_2}{\|\mathbf{u}\|} \\ &= \frac{u_1 v_1 + u_2 v_2}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.\end{aligned}$$

Since the denominator is nonzero,

$$\mathbf{u} \perp \mathbf{v} \iff u_1 v_1 + u_2 v_2 = 0.$$

The coordinate expression has emerged from the geometry. Only now do we give it a name.

Definition

For vectors in the plane, their *dot product* is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

In three-dimensional space,

$$(u_1, u_2, u_3) \cdot (v_1, v_2, v_3) = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

The preceding calculation proves

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0.$$

It proves more than the right-angle case. Because cosine is even and $\theta = |\beta - \alpha|$,

$$\cos(\beta - \alpha) = \cos\theta.$$

Therefore

$$\boxed{\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta}$$

for every pair of nonzero vectors in the plane.

Example

Take

$$\mathbf{u} = (3, 2), \qquad \mathbf{v} = (-2, 3).$$

Then

$$\mathbf{u} \cdot \mathbf{v} = 3(-2) + 2(3) = 0.$$

The coordinate calculation says that

$$\cos\theta = 0,$$

so the angle between the vectors is a right angle.

2.1.7 Why the Dot Product Can Be Expanded

The dot product was discovered from the cosine of the angle between two vectors. Before using it in longer calculations, we must verify how it behaves under addition and scalar multiplication.

Let

$$\mathbf{u} = (u_1, u_2), \qquad \mathbf{v} = (v_1, v_2), \qquad \mathbf{w} = (w_1, w_2).$$

Then

$$\begin{aligned}(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} &= (u_1 + w_1)v_1 + (u_2 + w_2)v_2 \\ &= (u_1v_1 + u_2v_2) + (w_1v_1 + w_2v_2) \\ &= \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v}.\end{aligned}$$

Thus the dot product distributes over vector addition. Similarly,

$$(\lambda\mathbf{u}) \cdot \mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v}).$$

The symmetry of multiplication of real numbers also gives

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}.$$

Theorem

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and a real number λ,

$$\begin{aligned}(\mathbf{u} + \mathbf{w}) \cdot \mathbf{v} &= \mathbf{u} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v}, \\ (\lambda\mathbf{u}) \cdot \mathbf{v} &= \lambda(\mathbf{u} \cdot \mathbf{v}), \\ \mathbf{u} \cdot \mathbf{v} &= \mathbf{v} \cdot \mathbf{u}.\end{aligned}$$

Moreover,

$$\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2.$$

The last identity follows immediately:

$$\mathbf{u} \cdot \mathbf{u} = u_1^2 + u_2^2 = \|\mathbf{u}\|^2.$$

These properties justify every later expansion involving expressions such as $(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})$. They are not formal rules borrowed from elsewhere; they follow from the coordinate definition of the dot product.

2.1.8 The Angle Between Two Vectors

The difference of inclination angles $\beta - \alpha$ is a directed turn and is understood modulo a full turn. The ordinary angle between two nonzero vectors is a different object: it is the smaller, unoriented separation between their directions.

Definition

For nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, their angle is the unique number

$$\theta \in [0, \pi]$$

satisfying

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

The interval $[0, \pi]$ chooses the smaller unoriented angle. Thus directions with inclinations $10°$ and $350°$ are separated by $20°$, not $340°$.

The definition agrees with the earlier geometric interpretation:

$$\theta = 0 \iff \text{same direction}, \qquad \theta = \frac{\pi}{2} \iff \text{perpendicular}, \qquad \theta = \pi \iff \text{opposite directions}.$$

The Cauchy–Schwarz inequality proved later guarantees that the quotient inside the cosine lies between -1 and 1.

2.1.9 The Cosine Theorem Emerges Immediately

Draw $\mathbf{u}$ and $\mathbf{v}$ from a common point. The displacement from the endpoint of $\mathbf{v}$ to the endpoint of $\mathbf{u}$ is

$$\mathbf{u} - \mathbf{v}.$$

Thus these three vectors determine a triangle with side lengths

$$\|\mathbf{u}\|, \qquad \|\mathbf{v}\|, \qquad \|\mathbf{u} - \mathbf{v}\|.$$

We calculate the square of the third side:

$$\begin{aligned}
\|\mathbf{u} - \mathbf{v}\|^2 &= (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) \\
&= \mathbf{u} \cdot \mathbf{u} - \mathbf{u} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v} \\
&= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u} \cdot \mathbf{v} \\
&= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta.
\end{aligned}$$

If

$$a = \|\mathbf{u}\|, \qquad b = \|\mathbf{v}\|, \qquad c = \|\mathbf{u} - \mathbf{v}\|,$$

then

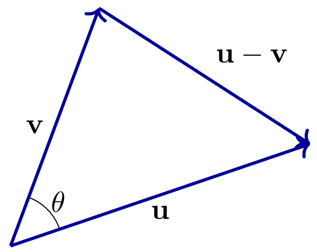
$$\boxed{c^2 = a^2 + b^2 - 2ab\cos\theta}.$$

Theorem

For a triangle with side lengths a, b, and c, where θ is the angle between the sides of lengths a and b,

$$c^2 = a^2 + b^2 - 2ab\cos\theta.$$

The cosine theorem is not inserted as an external formula. It is the coordinate expansion of the displacement between the endpoints of two arbitrary vectors.



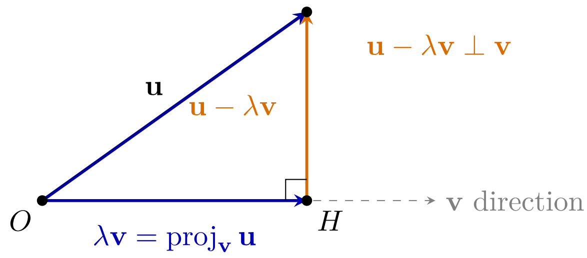
2.1.10 Projection Is Forced by the Right-Angle Condition

Let $\mathbf{v} \neq \mathbf{0}$. We seek a multiple $\lambda\mathbf{v}$ such that the remainder

$$\mathbf{u} - \lambda\mathbf{v}$$

is perpendicular to $\mathbf{v}$. Geometrically, this means decomposing $\mathbf{u}$ into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$:

$$\mathbf{u} = \lambda\mathbf{v} + (\mathbf{u} - \lambda\mathbf{v}).$$



The point H is chosen on the line spanned by $\mathbf{v}$. The vector from O to H is the parallel component $\lambda\mathbf{v}$, while the vector from H to the endpoint of $\mathbf{u}$ is the perpendicular remainder. The right-angle condition determines the value of λ.

The criterion just derived requires

$$(\mathbf{u} - \lambda\mathbf{v}) \cdot \mathbf{v} = 0.$$

Expanding gives

$$\mathbf{u} \cdot \mathbf{v} - \lambda(\mathbf{v} \cdot \mathbf{v}) = 0,$$

so

$$\lambda = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}.$$

Thus the parallel component is

$$\mathrm{proj}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v}.$$

The formula is the solution of a geometric requirement, not a rule to be memorised without explanation.

2.1.11 Orthogonal Decomposition and Its Consequences

The projection formula splits $\mathbf{u}$ into two geometrically different parts:

$$\mathbf{u} = \mathrm{proj}_{\mathbf{v}}\,\mathbf{u} + (\mathbf{u} - \mathrm{proj}_{\mathbf{v}}\,\mathbf{u})\,.$$

The first term is parallel to $\mathbf{v}$ and the second is perpendicular to $\mathbf{v}$. This is the orthogonal decomposition of $\mathbf{u}$ relative to the direction of $\mathbf{v}$.

Because the two components are perpendicular, the Pythagorean theorem gives

$$\boxed{\|\mathbf{u}\|^2 = \|\mathrm{proj}_{\mathbf{v}}\,\mathbf{u}\|^2 + \|\mathbf{u} - \mathrm{proj}_{\mathbf{v}}\,\mathbf{u}\|^2}.$$

Scalar component and vector projection

The signed scalar component of $\mathbf{u}$ in the direction of nonzero $\mathbf{v}$ is

$$\boxed{\mathrm{comp}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}}.$$

It measures how much signed length of $\mathbf{u}$ lies along the direction of $\mathbf{v}$. The vector projection is obtained by multiplying this number by the unit vector in that direction:

$$\mathrm{proj}_{\mathbf{v}}\,\mathbf{u} = \mathrm{comp}_{\mathbf{v}}\,\mathbf{u}\frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

The Cauchy–Schwarz inequality

The length of the parallel component cannot exceed the length of the entire vector:

$$|\mathrm{comp}_{\mathbf{v}}\,\mathbf{u}| \le \|\mathbf{u}\|.$$

Substitution gives

$$\boxed{|\mathbf{u}\cdot\mathbf{v}| \le \|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

Therefore

$$-1 \le \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \le 1,$$

which guarantees that the angle formula using arccos is meaningful.

The triangle inequality

The direct displacement $\mathbf{u}+\mathbf{v}$ cannot be longer than travelling first by $\mathbf{u}$ and then by $\mathbf{v}$. Algebraically,

$$\begin{aligned}\|\mathbf{u}+\mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2\\ &\le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\,\|\mathbf{v}\| + \|\mathbf{v}\|^2\\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2.\end{aligned}$$

Hence

$$\boxed{\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|}.$$

This is the algebraic form of the geometric fact that the straight segment is the shortest route between two points.

2.1.12 From the Plane to Three-Dimensional Vectors

Adding a third coordinate does not change the basic meaning of a vector. A vector still records a displacement, but now it records three independent coordinate changes:

$$\mathbf{v} = (v_1, v_2, v_3).$$

Addition and scalar multiplication remain componentwise, because successive changes in each coordinate direction still combine independently.

Applying the Pythagorean theorem first to the first two coordinate changes and then to the third gives

$$\boxed{\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}}.$$

The dot product extends in exactly the same way:

$$\boxed{\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3}.$$

Its geometric role is unchanged:

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta.$$

Consequently,

$$\boxed{\mathbf{u}\perp\mathbf{v} \iff \mathbf{u}\cdot\mathbf{v} = 0}.$$

Thus the dot product detects perpendicularity in both two and three dimensions.

A second product available in three dimensions

In three dimensions there is another natural question. Given two vectors $\mathbf{u}$ and $\mathbf{v}$, can we construct a new vector perpendicular to both of them?

Step 1: translate the geometric requirement into equations.
Let

$$\mathbf{u} = (u_1, u_2, u_3), \qquad \mathbf{v} = (v_1, v_2, v_3), \qquad \mathbf{n} = (n_1, n_2, n_3).$$

We require $\mathbf{n}$ to be perpendicular to both inputs:

$$\mathbf{n} \cdot \mathbf{u} = 0, \qquad \mathbf{n} \cdot \mathbf{v} = 0.$$

Therefore

$$\begin{aligned} u_1 n_1 + u_2 n_2 + u_3 n_3 &= 0, \\ v_1 n_1 + v_2 n_2 + v_3 n_3 &= 0. \end{aligned}$$

Step 2: choose one component so that the remaining system is easy.
Set

$$\Delta = u_1 v_2 - u_2 v_1.$$

Assume first that $\Delta \neq 0$, and choose

$$n_3 = \Delta.$$

The two equations reduce to the ordinary 2×2 system

$$\begin{aligned} u_1 n_1 + u_2 n_2 &= -u_3 \Delta, \\ v_1 n_1 + v_2 n_2 &= -v_3 \Delta. \end{aligned}$$

Step 3: eliminate n_2 and find n_1.
Multiply the first equation by v_2, the second by u_2, and subtract:

$$\begin{aligned} v_2(u_1 n_1 + u_2 n_2) &- u_2(v_1 n_1 + v_2 n_2) \\ &= -u_3 v_2 \Delta + u_2 v_3 \Delta. \end{aligned}$$

The n_2-terms cancel, so

$$\Delta n_1 = \Delta(u_2 v_3 - u_3 v_2).$$

Since $\Delta \neq 0$,

$$\boxed{n_1 = u_2 v_3 - u_3 v_2}.$$

Step 4: eliminate n_1 and find n_2.
Multiply the first equation by v_1, the second by u_1, and subtract:

$$\begin{aligned} v_1(u_1n_1 + u_2n_2) &- u_1(v_1n_1 + v_2n_2) \\ &= -u_3v_1\Delta + u_1v_3\Delta. \end{aligned}$$

The n_1-terms cancel, giving

$$-\Delta n_2 = \Delta(u_1v_3 - u_3v_1).$$

Hence

$$\boxed{n_2 = u_3v_1 - u_1v_3}.$$

Step 5: assemble the three components.
Together with

$$n_3 = u_1v_2 - u_2v_1,$$

we obtain

$$\boxed{\mathbf{n} = \big(u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1\big)}.$$

If the displayed Δ is zero but the vectors are not parallel, one of the other two coordinate pairs is nonzero. Repeating the same elimination with that pair gives the same cyclic formula.

Step 6: verify that the constructed vector really works.
For the first input vector, the terms cancel in three visible pairs:

$$\begin{aligned} \mathbf{n}\cdot\mathbf{u} &= (u_1u_2v_3 - u_1u_2v_3) \\ &\quad + (-u_1u_3v_2 + u_1u_3v_2) \\ &\quad + (u_2u_3v_1 - u_2u_3v_1) \\ &= 0. \end{aligned}$$

Likewise,

$$\begin{aligned} \mathbf{n}\cdot\mathbf{v} &= (u_2v_1v_3 - u_2v_1v_3) \\ &\quad + (-u_3v_1v_2 + u_3v_1v_2) \\ &\quad + (-u_1v_2v_3 + u_1v_2v_3) \\ &= 0. \end{aligned}$$

Thus $\mathbf{n} \perp \mathbf{u}$ and $\mathbf{n} \perp \mathbf{v}$.

The vector has therefore been derived from the two perpendicularity equations. We now give it a name:

$$\boxed{\mathbf{u}\times\mathbf{v} = \big(u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1\big)}.$$

Consequently,

$$\boxed{(\mathbf{u}\times\mathbf{v})\cdot\mathbf{u} = 0, \qquad (\mathbf{u}\times\mathbf{v})\cdot\mathbf{v} = 0}.$$

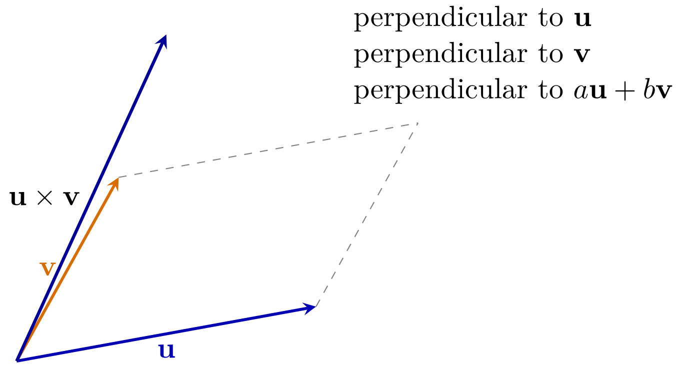
The statement is stronger. Every linear combination

$$a\mathbf{u} + b\mathbf{v}$$

is also perpendicular to $\mathbf{u} \times \mathbf{v}$, because

$$\begin{aligned}(\mathbf{u} \times \mathbf{v}) \cdot (a\mathbf{u} + b\mathbf{v}) &= a(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} + b(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} \\ &= a \cdot 0 + b \cdot 0 \\ &= 0.\end{aligned}$$

We do not yet turn this family of linear combinations into a theory of planes. That geometric construction belongs to the later chapter on lines and planes. For now, the important fact is simply that one cross product produces a vector perpendicular to every combination of its two inputs.



The cross product detects parallelism

If $\mathbf{u}$ and $\mathbf{v}$ are parallel, then one is a scalar multiple of the other. Let $\mathbf{v} = \lambda\mathbf{u}$. Substitution into the coordinate formula makes every component of the cross product vanish, so

$$\mathbf{u} \times \mathbf{v} = \mathbf{0}.$$

Conversely, if two nonzero vectors have zero cross product, then their directions are parallel. Hence

$$\boxed{\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = \mathbf{0}}$$

for nonzero vectors.

This gives two complementary geometric tests:

$$\boxed{\begin{array}{rcl} \mathbf{u} \cdot \mathbf{v} = 0 & \iff & \mathbf{u} \perp \mathbf{v}, \\ \mathbf{u} \times \mathbf{v} = \mathbf{0} & \iff & \mathbf{u} \parallel \mathbf{v} \quad (\mathbf{u}, \mathbf{v} \neq \mathbf{0}). \end{array}}$$

Order and orientation

There are two opposite directions perpendicular to both nonparallel vectors. To choose one consistently, we use the right-hand rule: curl the fingers of the right hand from $\mathbf{u}$ toward $\mathbf{v}$; the thumb indicates the direction of $\mathbf{u} \times \mathbf{v}$.

Reversing the order reverses the result:

$$\boxed{\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})}.$$

Thus the cross product is not commutative.

Its length satisfies

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta.$$

This formula explains the parallelism test: for parallel vectors, $\theta = 0$ or $\theta = \pi$, so the sine and therefore the cross product vanish. The later geometry chapters will use the same quantity when areas, lines, and planes are constructed.

Summary

The two vector products answer different geometric questions:

operation	output	main test
$\mathbf{u}\cdot\mathbf{v}$	number	$\mathbf{u}\perp\mathbf{v} \iff \mathbf{u}\cdot\mathbf{v}=0$
$\mathbf{u}\times\mathbf{v}$	vector	$\mathbf{u}\parallel\mathbf{v} \iff \mathbf{u}\times\mathbf{v}=\mathbf{0}$

Moreover, $\mathbf{u}\times\mathbf{v}$ is perpendicular to $\mathbf{u}$, to $\mathbf{v}$, and to every linear combination $a\mathbf{u}+b\mathbf{v}$. These facts are developed here as properties of vectors. They will become tools for constructing and analysing lines and planes only in the following chapter.

2.2 Changing the Basis: New Coordinates for the Same Vector

The preceding section developed the basic geometry and algebra of vectors: addition, scalar multiplication, linear combinations, length, angle, the dot product, parallelism, perpendicularity, and orthogonal projection. We now use that entire toolkit to study a new question: how the coordinate description of one fixed geometric vector changes when the basis is changed.

Up to this point every vector has been described with respect to the horizontal and vertical coordinate directions. That choice was natural because the Cartesian axes had already been constructed, but it was still only a choice. The geometric vector itself does not contain a preferred pair of coordinate axes.

We now ask a new question:

What numbers describe the same vector when we choose two different coordinate directions?

This is the problem of changing basis. The vector will remain fixed. Only the language used to describe it will change.

A basis is a pair of independent directions

Definition

Two nonparallel vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ form a basis of the plane. This means that every vector $\mathbf{u}$ can be written in one and only one way as

$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2.$$

The numbers c_1 and c_2 are called the coordinates of $\mathbf{u}$ in the basis

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2).$$

We write

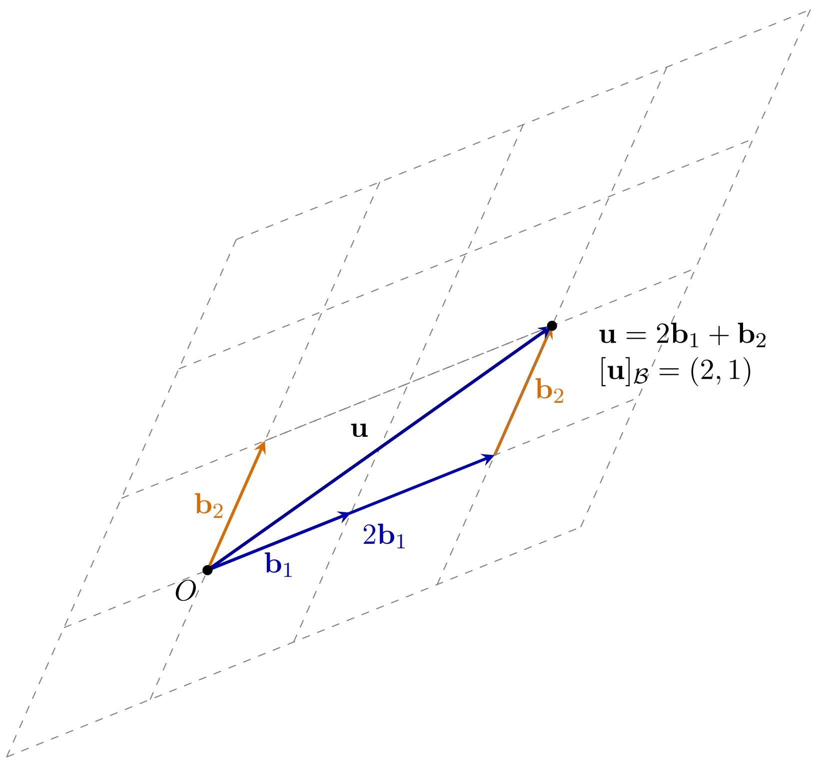
$$[\mathbf{u}]_{\mathcal{B}} = (c_1, c_2).$$

The basis vectors need not be perpendicular and need not have length one. The only forbidden situation is that they are parallel, because then both vectors point along the same line and cannot describe arbitrary displacements in the plane.

Geometrically, the equation

$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2$$

means the following. Starting at the origin, move first by $c_1\mathbf{b}_1$, then by $c_2\mathbf{b}_2$. The endpoint must coincide with the endpoint of $\mathbf{u}$.



The coefficients are read from the oblique parallelogram. They are not lengths of perpendicular shadows. They tell us how many signed copies of each basis vector are needed.

The same vector on two different grids

Let the ordinary Cartesian basis be

$$\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2), \qquad \mathbf{e}_1 = (1, 0), \qquad \mathbf{e}_2 = (0, 1).$$

Choose also the slanted basis

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2), \qquad \mathbf{b}_1 = (2, 1), \qquad \mathbf{b}_2 = (1, 2).$$

Consider the geometric vector

$$\mathbf{u} = (5, 4)$$

when it is described in the Cartesian basis.

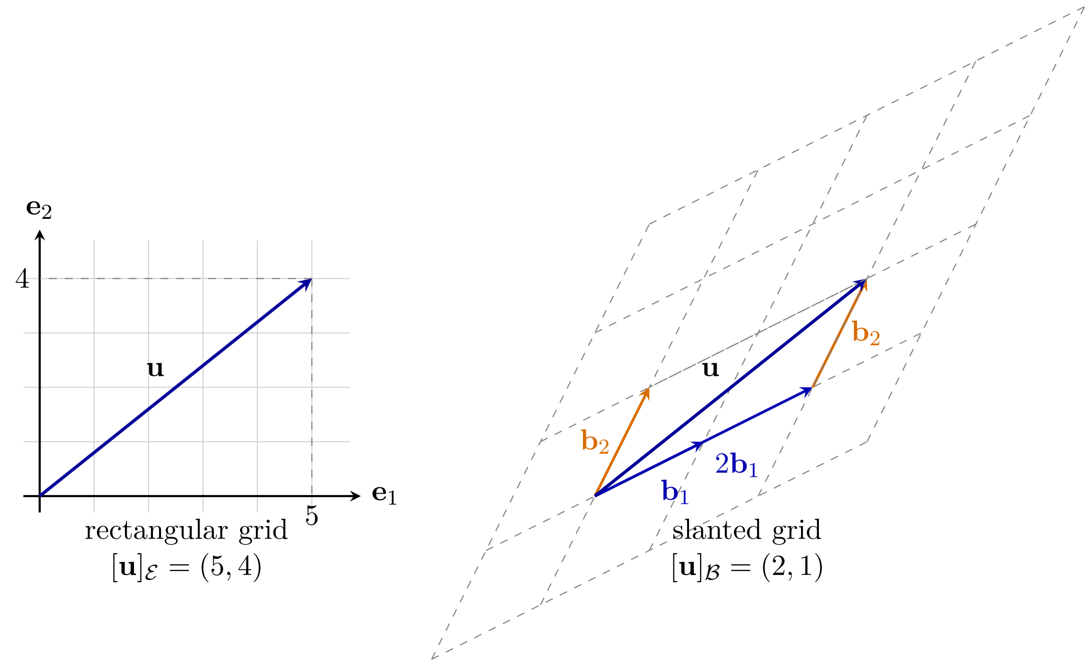
On the rectangular grid, the endpoint is found by moving five units in the $\mathbf{e}_1$ direction and four units in the $\mathbf{e}_2$ direction. Hence

$$[\mathbf{u}]_{\mathcal{E}} = (5, 4).$$

On the slanted grid, we must find numbers c_1, c_2 such that

$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2.$$

The two pictures below show the same arrow. Only the grid used to read its coordinates has changed.



On the rectangular grid we read the coordinates by projecting parallel to the coordinate axes. On the slanted grid we do the analogous construction, but the lines through the endpoint are drawn parallel to the other basis vector. This produces an oblique parallelogram rather than a rectangle.

Finding the new coordinates by a system of equations

The geometric picture tells us what must be true, but it does not yet calculate c_1 and c_2. Write the basis vectors in ordinary Cartesian coordinates:

$$\mathbf{b}_1 = (a_1, a_2), \qquad \mathbf{b}_2 = (b_1, b_2), \qquad \mathbf{u} = (x, y).$$

The equation

$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2$$

means equality of the two Cartesian components. Therefore

$$\boxed{\begin{aligned} a_1c_1 + b_1c_2 &= x, \\ a_2c_1 + b_2c_2 &= y. \end{aligned}}$$

This ordinary system of two linear equations is the complete algebraic content of changing from the Cartesian basis to the basis $\mathcal{B}$.

For our example,

$$\mathbf{b}_1 = (2, 1), \qquad \mathbf{b}_2 = (1, 2), \qquad \mathbf{u} = (5, 4),$$

so

$$\begin{aligned} 2c_1 + c_2 &= 5, \\ c_1 + 2c_2 &= 4. \end{aligned}$$

From the first equation,

$$c_2 = 5 - 2c_1.$$

Substituting into the second gives

$$c_1 + 2(5 - 2c_1) = 4,$$

so

$$-3c_1 = -6, \qquad c_1 = 2.$$

Then

$$c_2 = 5 - 2 \cdot 2 = 1.$$

Thus

$$\boxed{[\mathbf{u}]_{\mathcal{B}} = (2, 1)}.$$

The calculation agrees with the oblique parallelogram in the figure.

A general algebraic formula in the plane

The same system can be solved symbolically. Starting from

$$\begin{aligned} a_1c_1 + b_1c_2 &= x, \\ a_2c_1 + b_2c_2 &= y, \end{aligned}$$

multiply the first equation by b_2 and the second by b_1:

$$\begin{aligned} a_1b_2c_1 + b_1b_2c_2 &= xb_2, \\ a_2b_1c_1 + b_1b_2c_2 &= yb_1. \end{aligned}$$

Subtracting eliminates c_2:

$$(a_1b_2 - a_2b_1)c_1 = xb_2 - yb_1.$$

Similarly, eliminating c_1 gives

$$(a_1b_2 - a_2b_1)c_2 = a_1y - a_2x.$$

Hence

$$\boxed{c_1 = \frac{xb_2 - yb_1}{a_1b_2 - a_2b_1}, \qquad c_2 = \frac{a_1y - a_2x}{a_1b_2 - a_2b_1}}.$$

The denominator

$$a_1b_2 - a_2b_1$$

is nonzero precisely because the basis vectors are not parallel. If it were zero, the two chosen directions would collapse onto one line and the coordinates would not be uniquely determined.

Changing directly from one nonstandard basis to another

Suppose

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2) \qquad \text{and} \qquad \mathcal{C} = (\mathbf{c}_1, \mathbf{c}_2)$$

are two bases. Assume that

$$[\mathbf{u}]_{\mathcal{B}} = (p, q).$$

Then first reconstruct the geometric vector:

$$\mathbf{u} = p\mathbf{b}_1 + q\mathbf{b}_2.$$

If

$$\mathbf{b}_1 = (a_1, a_2), \qquad \mathbf{b}_2 = (b_1, b_2),$$

then

$$\mathbf{u} = (pa_1 + qb_1,\ pa_2 + qb_2).$$

Now seek numbers r, s such that

$$\mathbf{u} = r\mathbf{c}_1 + s\mathbf{c}_2.$$

Writing

$$\mathbf{c}_1 = (d_1, d_2), \qquad \mathbf{c}_2 = (e_1, e_2),$$

we solve

$$d_1 r + e_1 s = pa_1 + qb_1,$$
$$d_2 r + e_2 s = pa_2 + qb_2.$$

The solution (r, s) is exactly $[\mathbf{u}]_{\mathcal{C}}$.

Thus a change from one basis to another is not a mysterious new operation. It is two familiar steps:

$$\boxed{\text{old coordinates} \longrightarrow \text{the geometric vector} \longrightarrow \text{new coordinates.}}$$

Why perpendicular projection is special

For the Cartesian basis, the coordinate directions are perpendicular unit vectors. Therefore the horizontal and vertical coordinates can be read from ordinary perpendicular projections.

For a slanted basis, this is no longer true. Projecting $\mathbf{u}$ perpendicularly onto $\mathbf{b}_1$ and $\mathbf{b}_2$ produces two shadows, but those shadows generally do not add back to $\mathbf{u}$. The correct geometric construction uses lines parallel to the other basis vector, exactly as in the oblique grid above.

Remark

Later, after matrices have been introduced, the same systems of equations will be written much more compactly. Matrix notation will not create a new method; it will only abbreviate the elimination calculations performed here.

Summary

A basis is a pair of nonparallel coordinate directions. The same geometric vector may have different coordinate pairs in different bases. To find the coordinates in a slanted basis, write

$$\mathbf{u} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2$$

and solve the resulting two scalar equations. Rectangular coordinates use perpendicular projections; slanted coordinates use an oblique parallelogram whose sides are parallel to the basis vectors.

2.2.1 A Negative Coordinate Means Motion in the Opposite Direction

The example $[\mathbf{u}]_{\mathcal{B}} = (2, 1)$ used positive coefficients. That is convenient for drawing a parallelogram, but basis coordinates need not be positive.

Keep

$$\mathbf{b}_1 = (2, 1), \qquad \mathbf{b}_2 = (1, 2),$$

and consider

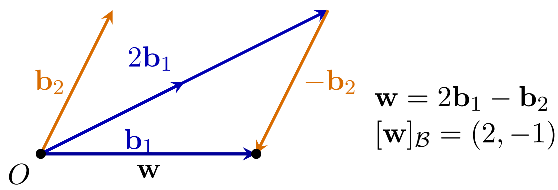
$$\mathbf{w} = 2\mathbf{b}_1 - \mathbf{b}_2.$$

Calculating with ordinary coordinate pairs gives

$$\mathbf{w} = 2(2, 1) - (1, 2) = (4, 2) - (1, 2) = (3, 0).$$

Therefore

$$[\mathbf{w}]_{\mathcal{E}} = (3, 0), \qquad [\mathbf{w}]_{\mathcal{B}} = (2, -1).$$



The coefficient -1 does not mean a negative length. It means one copy of $\mathbf{b}_2$ taken in the opposite direction. Basis coordinates are signed instructions for reconstructing the vector.

2.3 Beyond a Change of Basis: Polar Coordinates

The preceding section changed the basis of the plane. In every such change we kept two fixed, nonparallel vector directions and described a vector by the signed amounts of those two directions. The coordinate grid could become slanted, but it still consisted of two families of parallel straight lines.

There is a more radical possibility. We may keep the same Euclidean plane and describe a point by quantities that are not coefficients of two fixed basis vectors. Polar coordinates provide the first important example.

Remark

A change from one vector basis to another is a *linear* change of coordinates. Polar coordinates are not obtained from a new vector basis. Their coordinate curves are rays and circles, and the directions used to read the coordinates vary from point to point.

Two different questions about the same point

The Cartesian description of a point uses two signed coordinate values obtained from projections parallel to the coordinate directions:

$$P = (x, y).$$

This is natural when two fixed straight directions are important. We may instead ask:

1. How far is the point from the origin?
2. In which direction do we move from the origin to reach it?

These questions lead to the polar pair

$$(r, \theta),$$

where, for a point different from the origin,

$$r > 0, \qquad 0 \le \theta < 2\pi.$$

The number r is the Euclidean distance from the origin, and θ is the angle measured counterclockwise from the positive x-axis.

The origin is handled separately. In the usual non-unique polar notation every pair $(0, \theta)$ represents the origin. In this course we choose the single representative

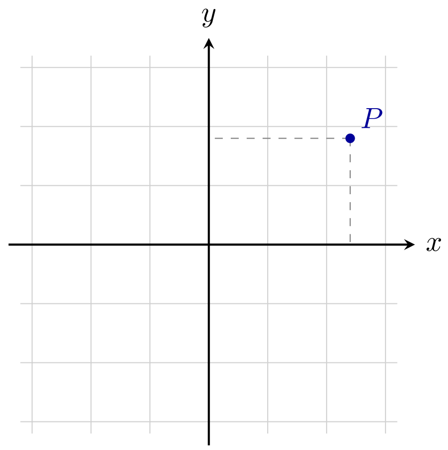
$$O = (0, 0)$$

so that every point has exactly one allowed polar pair. Thus

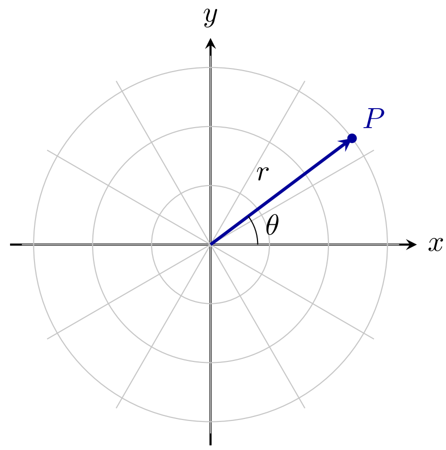
$$\mathcal{P} = \{(0, 0)\} \cup \big((0, \infty) \times [0, 2\pi)\big).$$

Rectangular and polar grids

The two descriptions below refer to the same Euclidean plane. What changes is the family of curves used to locate a point.



Cartesian grid
two families of parallel lines



polar grid
rays and concentric circles

In the Cartesian grid, keeping x fixed gives a vertical straight line and keeping y fixed gives a horizontal straight line. In the polar grid, keeping θ fixed gives a ray from the origin, while keeping r fixed gives a circle. This is the visible reason polar coordinates are not merely a slanted basis.

Converting between the descriptions

Let P have polar coordinates (r, θ). The right triangle determined by P gives

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta}.$$

Thus the position vector of P may be written as

$$\overrightarrow{OP} = r(\cos\theta, \sin\theta).$$

This resembles a scalar multiple of a unit vector, but the direction $(\cos\theta, \sin\theta)$ depends on the coordinate θ. There are not two fixed basis vectors whose constant coefficients are r and θ.

Conversely, if $(x, y) \neq (0, 0)$, then

$$\boxed{r = \sqrt{x^2 + y^2}}.$$

The polar angle is the unique number $\theta \in [0, 2\pi)$ satisfying

$$\boxed{\cos\theta = \frac{x}{r}, \qquad \sin\theta = \frac{y}{r}}.$$

These two equations determine both the quadrant and the exact direction. The single expression $\arctan(y/x)$ is insufficient because it loses quadrant information and is undefined when $x = 0$. In computation the correct angle is usually obtained with the two-argument function $\operatorname{atan2}(y, x)$, with a negative output shifted by 2π.

Example

For

$$(r,\theta) = \left(2, \frac{\pi}{3}\right),$$

we obtain

$$x = 2\cos\frac{\pi}{3} = 1, \qquad y = 2\sin\frac{\pi}{3} = \sqrt{3}.$$

Thus the same point is written as

$$P = (1, \sqrt{3})$$

in Cartesian coordinates and as

$$P = \left(2, \frac{\pi}{3}\right)$$

in polar coordinates.

Why the transformation is nonlinear

For a change of vector basis, multiplying a vector by λ multiplies each of its coordinates by λ, and the coordinates of a sum are obtained by adding coordinate pairs. Polar coordinates do not behave this way.

For example, the points with polar pairs

$$(1,0) \qquad \text{and} \qquad \left(1, \frac{\pi}{2}\right)$$

have Cartesian position vectors $(1,0)$ and $(0,1)$. Their vector sum is $(1,1)$, whose polar coordinates are

$$\left(\sqrt{2}, \frac{\pi}{4}\right),$$

not

$$\left(2, \frac{\pi}{2}\right).$$

Therefore polar coordinate pairs cannot be added componentwise to add vectors. The formulas

$$x = r\cos\theta, \qquad y = r\sin\theta$$

contain trigonometric functions and describe a genuinely nonlinear change of coordinates.

A circle becomes simple

Polar coordinates are useful when distance from the origin is the main geometric feature. Let

$$C_R = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = R^2\}, \qquad R > 0.$$

Substituting

$$x = r\cos\theta, \qquad y = r\sin\theta$$

gives

$$r^2\cos^2\theta + r^2\sin^2\theta = R^2.$$

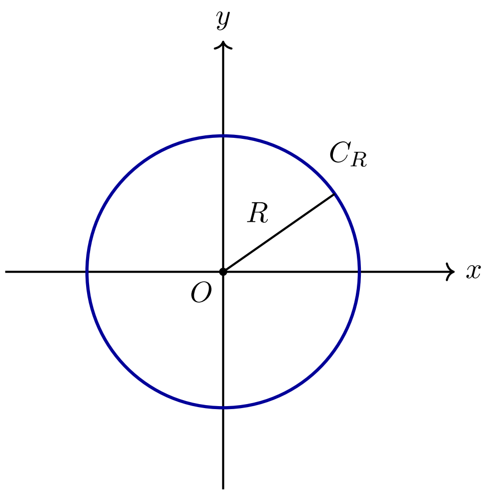
Since $\cos^2\theta + \sin^2\theta = 1$, this reduces to

$$\boxed{r = R}.$$

Thus

$$x^2 + y^2 = R^2 \qquad \text{and} \qquad r = R$$

describe the same geometric circle. The set has not changed; only the coordinate language has changed.



Summary

A vector basis uses fixed straight directions and leads to linear coordinate changes. Polar coordinates instead use a distance and an angle:

$$(r, \theta) \longleftrightarrow (r\cos\theta, r\sin\theta).$$

Their coordinate curves are circles and rays, their directions vary across the plane, and vector addition is not performed by adding polar coordinate pairs. This is our first example of a nonlinear coordinate system.

Chapter 3

Lines and Planes in the Plane and in Space

3.1 Lines in the Cartesian Plane

A line should not first appear as a formula to be recognised. We already have all the ingredients needed to construct it geometrically: a point tells us where to begin, and a nonzero vector tells us in which direction motion is allowed. The question is therefore:

Which points can be reached from one fixed point by moving only in one fixed direction?

Let

$$P_0 = (x_0, y_0)$$

be fixed and let

$$\mathbf{v} = (u, v) \neq \mathbf{0}$$

be a fixed direction vector. Every permitted displacement is a scalar multiple $t\mathbf{v}$. Starting from P_0 gives

$$P = P_0 + t\mathbf{v}, \qquad t \in \mathbb{R}.$$

Definition

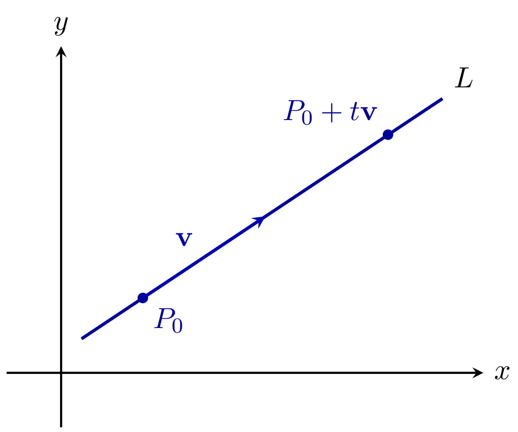
The line through P_0 with direction vector $\mathbf{v} \neq \mathbf{0}$ is

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{aligned} x &= x_0 + tu, \\ y &= y_0 + tv, \end{aligned} \qquad t \in \mathbb{R}.$$

The parameter has a direct meaning. At $t = 0$ we are at P_0; positive and negative values move in the two opposite orientations along the same direction. Changing the length of $\mathbf{v}$ only changes how quickly the parameter runs through the points, not the geometric line itself.



3.1.1 The same line described by a perpendicular direction

A direction vector tells us how to move along the line. There is a second, equally important description: choose a nonzero vector $\mathbf{n} = (a, b)$ perpendicular to $\mathbf{v}$. A point $P = (x, y)$ lies on the line precisely when the displacement $P - P_0$ is perpendicular to $\mathbf{n}$. The dot-product test from the previous chapter gives

$$\mathbf{n} \cdot (P - P_0) = 0.$$

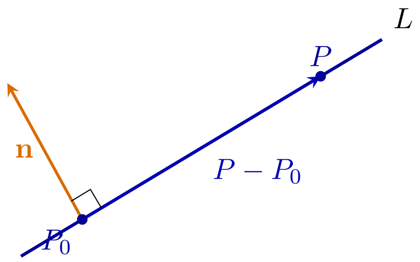
Written in coordinates,

$$a(x - x_0) + b(y - y_0) = 0.$$

After collecting the constant terms,

$$\boxed{ax + by = c}, \qquad c = ax_0 + by_0.$$

Thus the general linear equation is not a second, unrelated definition of a line. It records the same object by specifying a direction that no displacement along the line may contain.



3.1.2 Slope form as a special coordinate description

If the horizontal component u of the direction vector is nonzero, eliminate t from

$$x = x_0 + tu, \qquad y = y_0 + tv.$$

Since

$$t = \frac{x - x_0}{u},$$

we obtain

$$y - y_0 = \frac{v}{u}(x - x_0).$$

Therefore the slope is

$$\boxed{m = \frac{v}{u}},$$

and after collecting constants the equation takes the familiar form

$$y = mx + b.$$

If $u = 0$, the line is vertical and is described by $x = x_0$. The vector and normal descriptions treat this case without any exception; only slope notation fails.

Remark

The three forms answer different questions about the same set:

$P_0 + t\mathbf{v}$	how to move along the line,
$\mathbf{n} \cdot (P - P_0) = 0$	which displacements are allowed,
$y = mx + b$	how vertical position depends on horizontal position.

No representation is the line itself. The line is the geometric set of points that all these descriptions identify.

3.1.3 What the familiar Cartesian forms let us read

Once the same line has been described as

$$P = P_0 + t\mathbf{v}, \qquad \mathbf{n} \cdot (P - P_0) = 0, \qquad \text{or} \qquad y = mx + b,$$

we may choose the form best adapted to the question. The forms are not competing definitions; they expose different information about one geometric set.

Intersections with the coordinate axes

For a nonvertical line

$$y = mx + b,$$

setting $x = 0$ gives the intersection with the y-axis:

$$\boxed{(0, b)}.$$

If $m \neq 0$, setting $y = 0$ gives

$$x = -\frac{b}{m},$$

so the intersection with the x-axis is

$$\boxed{\left(-\frac{b}{m}, 0\right)}.$$

This illustrates a general habit: to find an intersection with a known set, impose both defining conditions at once.

Example

For

$$y = -\frac{1}{2}x + 3,$$

the y-axis intersection is $(0, 3)$. Setting $y = 0$ gives $x = 6$, so the x-axis intersection is $(6, 0)$.

The line through two points

Let

$$P = (x_1, y_1), \qquad Q = (x_2, y_2), \qquad P \neq Q.$$

The displacement

$$Q - P = (x_2 - x_1, y_2 - y_1)$$

is a direction vector, so the line is immediately

$$\boxed{L = \{P + t(Q - P) : t \in \mathbb{R}\}.}$$

This formula treats vertical and nonvertical lines uniformly.

If $x_1 \neq x_2$, the horizontal component of $Q - P$ is nonzero and the slope form may be recovered:

$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

and

$$\boxed{y - y_1 = m(x - x_1).}$$

If $x_1 = x_2$, no real slope exists and the line is simply

$$\boxed{x = x_1.}$$

The exceptional case belongs only to slope notation, not to the geometric line.

Parallel lines in slope notation

Two nonvertical lines

$$y = m_1 x + b_1, \qquad y = m_2 x + b_2$$

have direction vectors $(1, m_1)$ and $(1, m_2)$. They have the same direction precisely when

$$m_1 = m_2.$$

If also $b_1 = b_2$, the equations describe the same line. If $b_1 \neq b_2$, the lines are distinct and parallel. Thus

$$\boxed{L_1 \parallel L_2 \iff m_1 = m_2 \text{ and } b_1 \neq b_2}$$

for distinct nonvertical lines.

Remark

The slope formulas are efficient coordinate consequences of the vector construction. They should be used when they simplify a question, but they should not hide the more general descriptions, which also handle vertical lines and extend unchanged to three-dimensional space.

3.1.4 Vector tests behind the slope rules

Let two lines have nonzero direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Their geometric directions are parallel precisely when one direction vector is a nonzero scalar multiple of the other:

$$\boxed{\mathbf{v}_1 \parallel \mathbf{v}_2 \iff \mathbf{v}_2 = \lambda \mathbf{v}_1 \text{ for some } \lambda \neq 0.}$$

In three coordinates the same condition will be detected by a zero cross product. In the plane, the scalar-multiple criterion is already sufficient.

The directions are perpendicular precisely when

$$\boxed{\mathbf{v}_1 \perp \mathbf{v}_2 \iff \mathbf{v}_1 \cdot \mathbf{v}_2 = 0.}$$

These tests do not require a slope and therefore include vertical directions without a separate convention.

For nonvertical lines with slopes m_1, m_2, convenient direction vectors are

$$\mathbf{v}_1 = (1, m_1), \qquad \mathbf{v}_2 = (1, m_2).$$

Their dot product is

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 1 + m_1 m_2.$$

Consequently,

$$\mathbf{v}_1 \perp \mathbf{v}_2 \iff 1 + m_1 m_2 = 0 \iff \boxed{m_1 m_2 = -1}.$$

Thus the familiar slope rule is not an independent fact. It is the coordinate form of the dot-product test for the special direction vectors $(1, m)$.

3.1.5 The nearest point on a line

Let

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}, \qquad \mathbf{v} \neq \mathbf{0},$$

and let P be a point not necessarily on L. The displacement

$$P - P_0$$

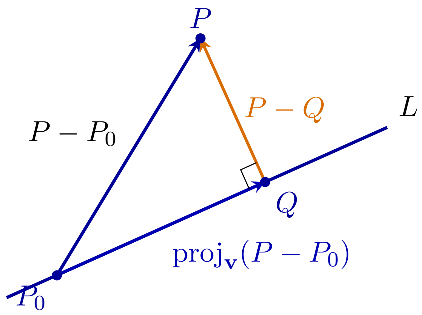
can be decomposed into a component parallel to $\mathbf{v}$ and a component perpendicular to $\mathbf{v}$. The parallel component is

$$\operatorname{proj}_{\mathbf{v}}(P - P_0) = \frac{(P - P_0) \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Starting from P_0 and moving by this component reaches the foot of the perpendicular:

$$\boxed{Q = P_0 + \operatorname{proj}_{\mathbf{v}}(P - P_0).}$$

The point Q lies on the line, and $P - Q$ is perpendicular to the line. It is therefore the shortest displacement from P to any point of L.



Hence

$$\boxed{d(P, L) = \|(P - P_0) - \operatorname{proj}_{\mathbf{v}}(P - P_0)\|\,.}$$

3.1.6 Distance from the normal equation

Suppose the same line is written as

$$ax + by = c, \qquad (a, b) \neq (0, 0),$$

with normal vector

$$\mathbf{n} = (a, b).$$

For $P = (x_P, y_P)$ and any fixed P_0 on the line,

$$\mathbf{n} \cdot (P - P_0) = ax_P + by_P - c.$$

The signed scalar component of $P - P_0$ in the normal direction is therefore

$$\frac{ax_P + by_P - c}{\sqrt{a^2 + b^2}}.$$

Distance is nonnegative, so

$$\boxed{d(P, L) = \frac{|ax_P + by_P - c|}{\sqrt{a^2 + b^2}}.}$$

This formula is not a new rule detached from projection. It is the length of the normal component of the displacement from the line to the point.

3.2 Lines in Three-Dimensional Space

A point in three-dimensional Cartesian space has coordinates

$$P = (x, y, z) \in \mathbb{R}^3.$$

A line is determined by one point and one nonzero direction.

Definition

Let

$$P_0 = (x_0, y_0, z_0)$$

and let

$$\mathbf{v} = (a, b, c) \neq (0, 0, 0).$$

The line through P_0 with direction vector $\mathbf{v}$ is

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}.$$

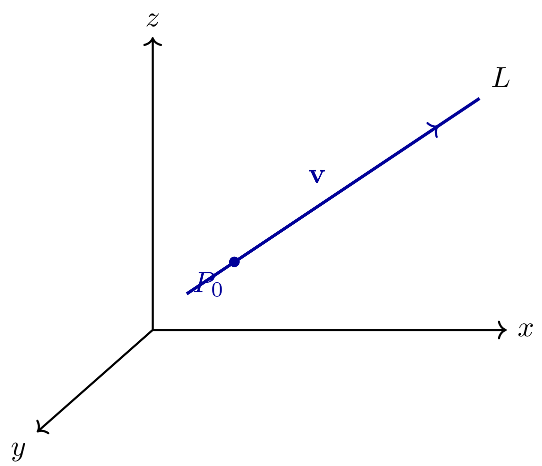
Equivalently,

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c).$$

In coordinates this becomes the system

$$\begin{cases} x = x_0 + at, \\ y = y_0 + bt, \\ z = z_0 + ct, \end{cases} \qquad t \in \mathbb{R}.$$

As t varies, the point moves along the line. Positive and negative values of t describe the two opposite directions along the same geometric line.



3.2.1 The Line Through Two Points in Space

Let

$$P = (x_1, y_1, z_1), \qquad Q = (x_2, y_2, z_2)$$

be distinct points. Their difference

$$Q - P = (x_2 - x_1,\, y_2 - y_1,\, z_2 - z_1)$$

is a nonzero direction vector of the line through them.

Theorem

The unique line through two distinct points $P, Q \in \mathbb{R}^3$ is

$$L = \{P + t(Q - P) : t \in \mathbb{R}\}.$$

In coordinates,

$$\begin{cases} x = x_1 + t(x_2 - x_1), \\ y = y_1 + t(y_2 - y_1), \\ z = z_1 + t(z_2 - z_1). \end{cases}$$

Example

For

$$P = (1, 0, 2), \qquad Q = (3, 4, 1),$$

a direction vector is

$$Q - P = (2, 4, -1).$$

Therefore the line through P and Q is

$$(x, y, z) = (1, 0, 2) + t(2, 4, -1),$$

or

$$\begin{cases} x = 1 + 2t, \\ y = 4t, \\ z = 2 - t. \end{cases}$$

3.2.2 Symmetric Form

If a, b, and c are all nonzero, the parameter can be eliminated from

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct.$$

This gives

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

If one component of the direction vector is zero, the corresponding coordinate is constant. For example, if $c = 0$, then $z = z_0$, while the remaining two coordinates may still be written in symmetric form.

3.2.3 Parallel and Intersecting Lines in Space

Consider

$$L_1 = P_1 + t\mathbf{v}_1, \qquad L_2 = P_2 + s\mathbf{v}_2.$$

The lines are parallel when their direction vectors are proportional:

$$\mathbf{v}_2 = \lambda \mathbf{v}_1$$

for some nonzero λ.

To test whether two nonparallel lines intersect, one solves

$$P_1 + t\mathbf{v}_1 = P_2 + s\mathbf{v}_2$$

for the two parameters t and s. This is a system of three scalar equations. It may have one solution or no solution.

Definition

Two lines in $\mathbb{R}^3$ are *skew* when they are neither parallel nor intersecting.

Skew lines have no analogue in the plane. In two dimensions, two distinct nonparallel lines must intersect; in three dimensions, one line may pass above or beside another without meeting it.

Summary

A line in $\mathbb{R}^3$ is described by

$$L = \{P_0 + t\mathbf{v} : t \in \mathbb{R}\}, \qquad \mathbf{v} \neq \mathbf{0}.$$

Two distinct points determine the direction $Q - P$. Parallel lines have proportional direction vectors, while nonparallel lines may either intersect or be skew.

3.2.4 Vector tests for lines in space

Let

$$L_1 = P_1 + t\mathbf{v}_1, \qquad L_2 = P_2 + s\mathbf{v}_2.$$

The directions are parallel precisely when

$$\boxed{\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{0}.}$$

This is the three-dimensional parallelism test established in the vector chapter. It determines whether the directions agree, but it does not by itself distinguish identical lines from distinct parallel lines. For that, one must also check whether $P_2 - P_1$ is parallel to the common direction.

The direction vectors are perpendicular precisely when

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 0.$$

However, in space this alone does not imply that the geometric lines are perpendicular. Two lines are perpendicular only when they both intersect and their directions form a right angle. Without intersection they may be skew even though their direction vectors are perpendicular.

Example

Consider

$$L_1 = (0, 0, 0) + t(1, 0, 0)$$

and

$$L_2 = (0, 1, 1) + s(0, 1, 0).$$

Their direction vectors satisfy

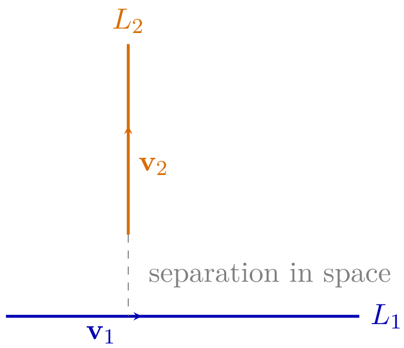
$$(1, 0, 0) \cdot (0, 1, 0) = 0.$$

Yet a point on L_1 has third coordinate 0, while every point on L_2 has third coordinate 1. The lines cannot intersect. They have perpendicular directions but are skew, not perpendicular lines.

To decide whether two nonparallel lines intersect, solve

$$P_1 + t\mathbf{v}_1 = P_2 + s\mathbf{v}_2.$$

This produces three scalar equations for the two parameters. A common pair (t, s) gives the intersection point; an inconsistent system proves that the lines are skew.



3.3 Planes in Three-Dimensional Space

A line was produced by allowing one real parameter:

$$P = P_0 + t\mathbf{v}.$$

Only one independent direction of motion was available. We now change the requirement:

What geometric set is obtained when motion from one fixed point is allowed in two independent directions?

Let $P_0 \in \mathbb{R}^3$ be fixed and let $\mathbf{u}, \mathbf{v}$ be nonparallel vectors. Every permitted displacement is a linear combination

$$s\mathbf{u} + t\mathbf{v}.$$

Starting at P_0 gives

$$\boxed{\Pi = \{P_0 + s\mathbf{u} + t\mathbf{v} : s, t \in \mathbb{R}\}.}$$

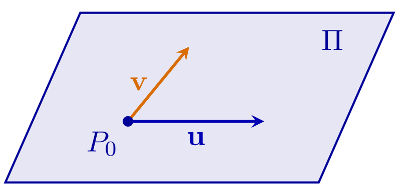
One parameter produced a line; two independent parameters produce a plane. If the two vectors were parallel, both parameters would still move us along only one direction and no plane would be created.

In coordinates, for

$$P_0 = (x_0, y_0, z_0), \quad \mathbf{u} = (u_1, u_2, u_3), \quad \mathbf{v} = (v_1, v_2, v_3),$$

we obtain

$$\begin{cases} x = x_0 + su_1 + tv_1, \\ y = y_0 + su_2 + tv_2, \\ z = z_0 + su_3 + tv_3. \end{cases}$$



3.3.1 Why a normal vector appears

The vector chapter proved that

$$(\mathbf{u} \times \mathbf{v}) \cdot (a\mathbf{u} + b\mathbf{v}) = 0$$

for every pair of real numbers a, b. Therefore

$$\mathbf{n} = \mathbf{u} \times \mathbf{v}$$

is perpendicular to every displacement allowed in the plane. This is not an extra definition imposed on the picture: it is forced by the two generating directions.

A point P belongs to the plane exactly when the displacement $P - P_0$ is one of those allowed combinations. Consequently it must satisfy

$$\boxed{\mathbf{n} \cdot (P - P_0) = 0.}$$

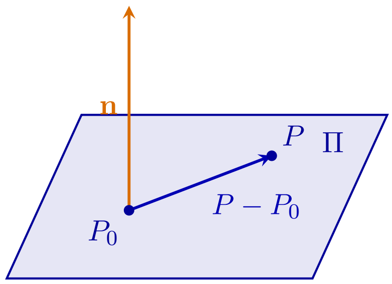
For $\mathbf{n} = (a, b, c)$ this becomes

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

or, after collecting constants,

$$\boxed{ax + by + cz = d.}$$

The parametric and normal equations describe the same plane from two different points of view: one lists allowed motions, while the other states which component of motion must vanish.



3.3.2 A plane through three points

Let P, Q, R be three points that do not lie on one line. The displacements

$$\mathbf{u} = Q - P, \qquad \mathbf{v} = R - P$$

are not parallel. Hence they provide the two independent directions required to construct a plane:

$$\boxed{\Pi = \{P + s(Q - P) + t(R - P) : s, t \in \mathbb{R}\}.}$$

Their cross product

$$\mathbf{n} = (Q - P) \times (R - P)$$

is nonzero and perpendicular to both directions, hence to every displacement in the plane. It therefore supplies the normal equation.

Example

For

$$P = (1, 0, 0), \qquad Q = (0, 1, 0), \qquad R = (0, 0, 1),$$

we obtain

$$Q - P = (-1, 1, 0), \qquad R - P = (-1, 0, 1),$$

and

$$(Q - P) \times (R - P) = (1, 1, 1).$$

Using the point P gives

$$(1, 1, 1) \cdot \big((x, y, z) - (1, 0, 0)\big) = 0,$$

so

$$x + y + z = 1.$$

3.3.3 Relative positions of planes

Let

$$\Pi_1 : \mathbf{n}_1 \cdot P = d_1, \qquad \Pi_2 : \mathbf{n}_2 \cdot P = d_2.$$

Their relative position is controlled by the normal directions.

If $\mathbf{n}_2 = \lambda \mathbf{n}_1$, the planes have the same orientation. They are either identical or distinct and parallel, depending on the constants. If the normal vectors are not parallel, the planes meet in a line.

The angle between two planes is measured through their normals. In particular,

$$\boxed{\Pi_1 \perp \Pi_2 \iff \mathbf{n}_1 \cdot \mathbf{n}_2 = 0.}$$

When two nonparallel planes intersect, a direction vector of their intersection must be perpendicular to both normals. The cross product therefore gives

$$\boxed{\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2.}$$

A point of the intersection is then found by solving the two plane equations.

3.3.4 Distance from a point to a plane

Let P be any point and let P_0 be a fixed point on the plane. The shortest motion from P to the plane must follow the normal direction. Only the component of $P - P_0$ parallel to $\mathbf{n}$ contributes to that distance. Hence

$$\boxed{d(P, \Pi) = \frac{|\mathbf{n} \cdot (P - P_0)|}{\|\mathbf{n}\|}.}$$

For $ax + by + cz = d$ and $P = (x_P, y_P, z_P)$ this becomes

$$\boxed{d(P, \Pi) = \frac{|ax_P + by_P + cz_P - d|}{\sqrt{a^2 + b^2 + c^2}}.}$$

The formula is therefore the length of an orthogonal projection, not an isolated rule.

Summary

A line is generated by one independent direction,

$$P = P_0 + t\mathbf{v},$$

whereas a plane is generated by two,

$$P = P_0 + s\mathbf{u} + t\mathbf{v}.$$

The cross product of the two generating directions produces a normal vector, and the dot product turns perpendicularity into the equation

$$\mathbf{n} \cdot (P - P_0) = 0.$$

Thus the parametric and normal descriptions arise from the same geometric construction.

Chapter 4

Conic Sections and Quadric Surfaces

4.1 Conic Sections: From Geometric Conditions to Equations

The previous chapter studied sets determined by first-degree equations. A line could be constructed from a point and a direction, or characterised by a perpendicularity condition. We now ask a different kind of question:

What curves arise when distances between moving and fixed geometric objects are constrained?

Distance in Cartesian coordinates contains squares. When a geometric distance condition is written algebraically and square roots are removed, second-degree terms appear naturally. Quadratic equations are therefore not introduced as an arbitrary next list of formulas; they arise from simple Euclidean questions.

4.1.1 A circle: one fixed point and one fixed distance

Fix a centre $C = (h, k)$ and a number $r > 0$. A point $P = (x, y)$ belongs to the circle precisely when

$$d(P, C) = r.$$

Using the Euclidean distance formula,

$$\sqrt{(x-h)^2 + (y-k)^2} = r.$$

Both sides are nonnegative, so squaring preserves the condition:

$$\boxed{(x-h)^2 + (y-k)^2 = r^2.}$$

The familiar quadratic equation has therefore been derived from the geometric definition of a circle.

4.1.2 A parabola: equal distance from a point and a line

Fix a point F, called the focus, and a line D, called the directrix. A parabola is the set of points whose distance from F equals their distance from D.

Choose coordinates so that

$$F = (0, p), \qquad D: \; y = -p, \qquad p > 0.$$

For $P = (x, y)$, the two distances are

$$d(P, F) = \sqrt{x^2 + (y-p)^2}$$

and

$$d(P, D) = |y + p|.$$

On the resulting parabola $y \geq -p$, so $|y + p| = y + p$. The defining equality gives

$$\sqrt{x^2 + (y-p)^2} = y + p.$$

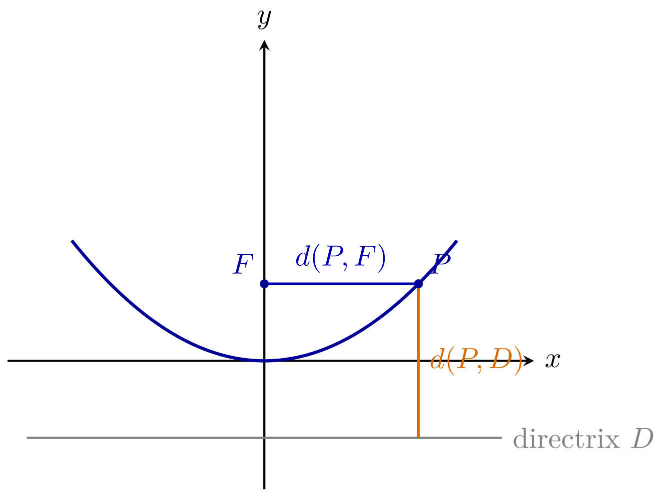
Squaring and cancelling equal terms,

$$x^2 + y^2 - 2py + p^2 = y^2 + 2py + p^2,$$

so

$$\boxed{x^2 = 4py.}$$

The equation is not the definition. It is the Cartesian form of the equal-distance condition.



4.1.3 An ellipse: a fixed sum of two distances

Fix two foci

$$F_1 = (-c, 0), \qquad F_2 = (c, 0), \qquad c > 0,$$

and require

$$d(P, F_1) + d(P, F_2) = 2a, \qquad a > c.$$

The inequality $a > c$ is necessary: the prescribed sum must be larger than the distance $2c$ between the foci.

For $P = (x, y)$ write

$$r_1 = \sqrt{(x+c)^2 + y^2}, \qquad r_2 = \sqrt{(x-c)^2 + y^2}.$$

The condition is $r_1 + r_2 = 2a$. Isolate $r_1 = 2a - r_2$ and square:

$$(x+c)^2 + y^2 = 4a^2 - 4ar_2 + (x-c)^2 + y^2.$$

After cancellation,

$$r_2 = a - \frac{c}{a}x.$$

Squaring once more gives

$$(x-c)^2 + y^2 = \left(a - \frac{c}{a}x\right)^2.$$

The linear terms cancel, leaving

$$\left(1 - \frac{c^2}{a^2}\right)x^2 + y^2 = a^2 - c^2.$$

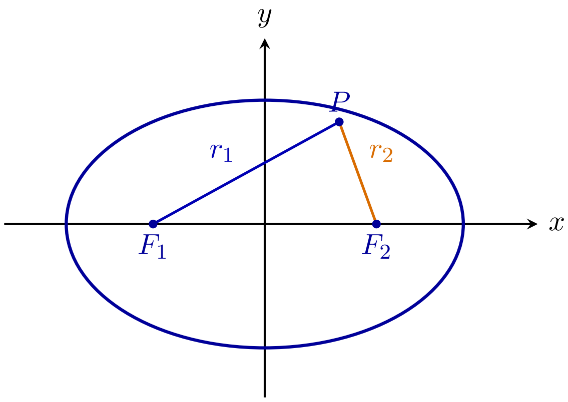
Define

$$b^2 = a^2 - c^2 > 0.$$

Then

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.}$$

The bounded oval is therefore not introduced by its equation. It is forced by the requirement that the total journey from one focus to the moving point and then to the other focus remain constant.



4.1.4 A hyperbola: a fixed difference of two distances

Keep the same foci, but now require

$$|d(P, F_1) - d(P, F_2)| = 2a, \qquad 0 < a < c.$$

The absolute value allows either focus to be the nearer one, producing two branches. On the right branch we have

$$r_1 - r_2 = 2a.$$

Thus $r_1 = 2a + r_2$. Squaring, isolating the remaining square root, and squaring a second time gives

$$\left(\frac{c^2}{a^2} - 1\right) x^2 - y^2 = c^2 - a^2.$$

Set

$$b^2 = c^2 - a^2 > 0.$$

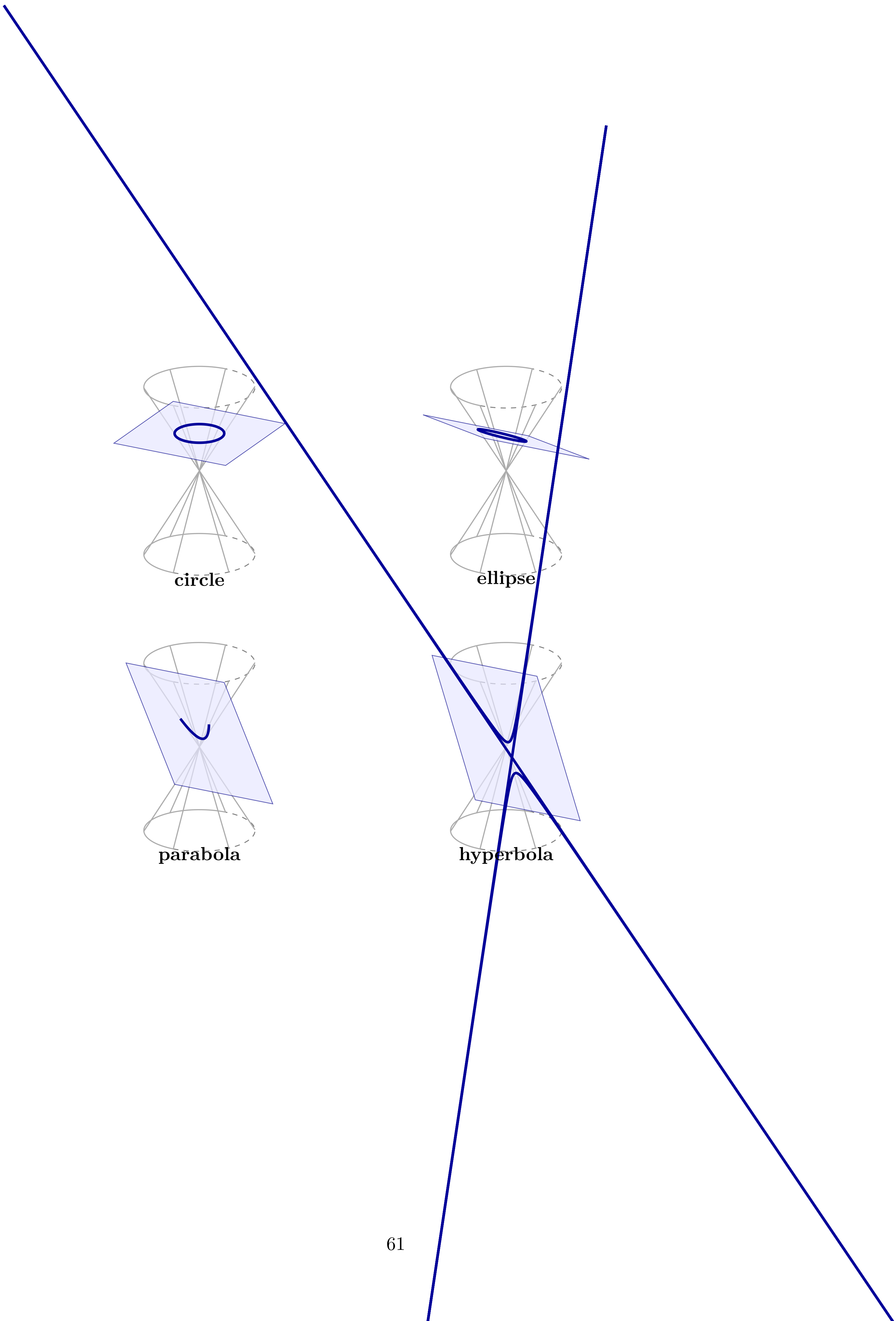
Then

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.}$$

Changing a constant sum into a constant difference changes a bounded curve into two unbounded branches. The new geometry is produced by changing the demand, not by choosing an unrelated formula.

4.1.5 Why these curves form one family

The four distance constructions also arise from one spatial experiment: intersect a double cone by a plane and vary the inclination of the cutting plane.


circle
ellipse
parabola
hyperbola

The four panels now use the same cone and the same algebraic construction. A horizontal plane gives a circle. Tilting the plane while it still meets only one nappe gives an ellipse. At the limiting inclination, when the plane is parallel to a generator, the intersection is a parabola. A steeper plane meets both nappes and produces a hyperbola. Thus the family changes because one geometric requirement on the cutting plane is varied while the cone remains fixed.

Remark

The rest of the chapter studies how these geometric sets appear in Cartesian equations. Standard forms, translations, rotations, and discriminants are tools for recognising the object already present in the equation. They do not replace the geometric definition of the curve.

4.1.6 From construction to recognition

The geometric definitions above answer the question *what the curves are*. A different problem begins when an equation is given first:

Which geometric curve, if any, is hidden in this algebraic condition?

When no mixed term is present, an equation of the form

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

can be analysed by completing squares. The signs of the squared terms then have a direct geometric meaning:

same nonzero sign	ellipse type or a degenerate case,
opposite signs	hyperbola type or intersecting lines,
one quadratic direction absent	parabola type or a degenerate case.

This reasoning is already available with ordinary algebra.

The mixed term Bxy indicates that the chosen coordinate axes may not follow the natural symmetry directions of the curve. Rotating the axes can remove that term. The quantity

$$B^2 - 4AC$$

packages the sign information in a way that survives such rotations. We will use its sign as a recognition criterion, but its full structural explanation belongs to the later study of matrices and quadratic forms.

Remark

There is no contradiction in using a theorem before giving its deepest possible explanation, provided its status is stated honestly. Here the geometric curves have already been constructed; the discriminant is introduced as an efficient recognition tool whose conceptual compression will be revisited later.

Algebraic Classification of Second-Degree Curves

Lines arise from equations in which the coordinates occur only to the first power. The next natural class consists of equations in which terms of total degree two are allowed.

Definition

A *second-degree equation* in the Cartesian plane has the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where $A, B, C, D, E, F \in \mathbb{R}$ and at least one of A, B, or C is nonzero. Its solution set is the collection of all points $(x, y) \in \mathbb{R}^2$ whose coordinates satisfy the equation.

The terms

$$Ax^2 + Bxy + Cy^2$$

form the quadratic part. The remaining terms affect the position and size of the curve, while the quadratic part largely determines its geometric type.

A First Classification

For the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

define

$$\Delta = B^2 - 4AC.$$

Theorem

For a non-degenerate real conic, the sign of Δ gives the following classification:

$$\begin{array}{r|c} \Delta < 0 & \text{ellipse type, including circles,} \\ \Delta = 0 & \text{parabola type,} \\ \Delta > 0 & \text{hyperbola type.} \end{array}$$

This classification is preserved by rotations and translations of the coordinate system. A rotation can remove the mixed term Bxy, while a translation can move the centre or vertex to a simpler position.

Remark

The sign of Δ identifies the conic type only after degenerate cases are separated. A second-degree equation may also describe a single point, no real points, one line, or two lines. These possibilities are discussed below.

Circle

A circle with centre (h, k) and radius $r > 0$ is described by

$$(x - h)^2 + (y - k)^2 = r^2.$$

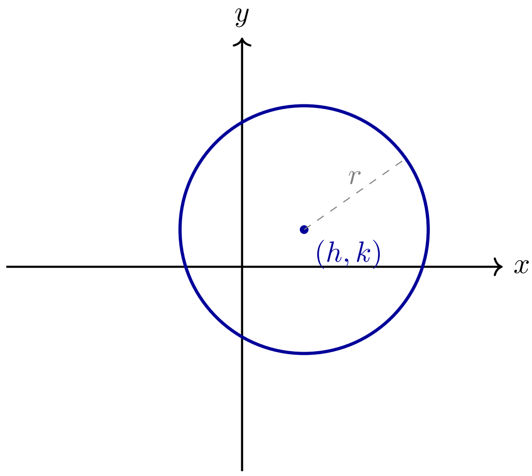
After expansion,

$$x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0.$$

Here $A = C = 1$ and $B = 0$, so

$$\Delta = 0^2 - 4 \cdot 1 \cdot 1 = -4 < 0.$$

A circle is therefore a special ellipse-type conic in which the coefficients of x^2 and y^2 are equal and there is no mixed term after a suitable rotation.



Ellipse

The standard ellipse centred at the origin has equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > 0, \quad b > 0.$$

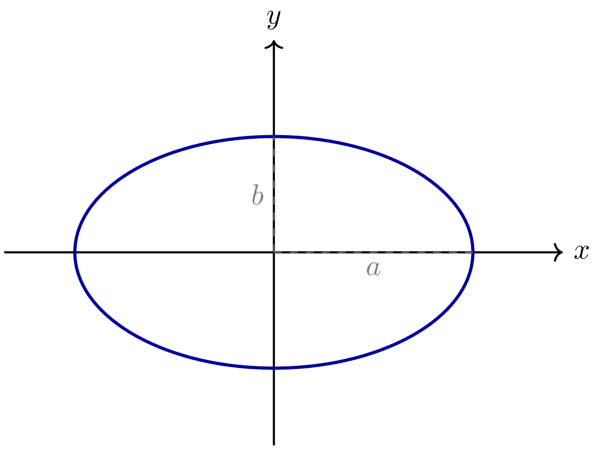
Equivalently,

$$b^2x^2 + a^2y^2 - a^2b^2 = 0.$$

Thus $A = b^2$, $B = 0$, and $C = a^2$, so

$$\Delta = -4a^2b^2 < 0.$$

The parameters a and b are the semi-axis lengths. When $a = b$, the ellipse becomes a circle.



Parabola

A standard vertical parabola has equation

$$y = ax^2, \qquad a \neq 0,$$

or equivalently

$$ax^2 - y = 0.$$

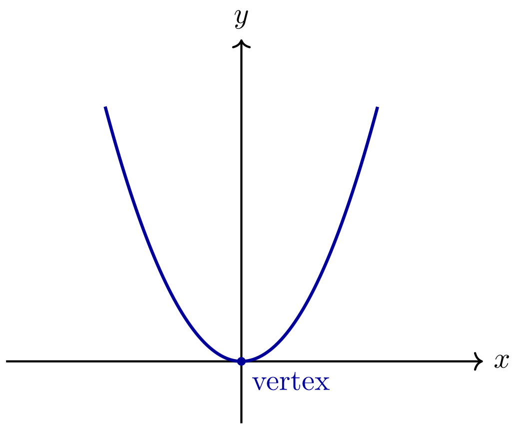
Here $A = a$, $B = 0$, and $C = 0$, hence

$$\Delta = 0.$$

The sign of a determines whether the parabola opens upward or downward. A translated form is

$$y - k = a(x - h)^2,$$

whose vertex is (h, k).



Hyperbola

A standard hyperbola centred at the origin has equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \qquad a > 0, \quad b > 0.$$

Equivalently,

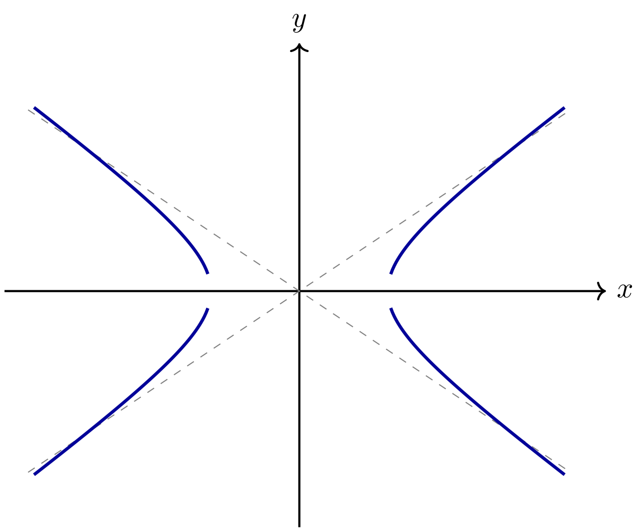
$$b^2x^2 - a^2y^2 - a^2b^2 = 0.$$

Here $A = b^2$, $B = 0$, and $C = -a^2$, so

$$\Delta = 4a^2b^2 > 0.$$

Its asymptotes are

$$y = \frac{b}{a}x, \qquad y = -\frac{b}{a}x.$$



The Mixed Term and Rotation

The term Bxy indicates that the natural axes of the conic may be rotated relative to the chosen Cartesian axes. For example,

$$x^2 + 2xy + y^2 = 1$$

can be written as

$$(x + y)^2 = 1,$$

so it is not a non-degenerate ellipse: it describes the two parallel lines

$$x + y = 1, \qquad x + y = -1.$$

For a genuine rotated conic, one introduces new coordinates (u, v) obtained by a rotation through an angle θ:

$$x = u\cos\theta - v\sin\theta, \qquad y = u\sin\theta + v\cos\theta.$$

Choosing θ appropriately removes the uv term. When $A \neq C$, one may choose θ so that

$$\tan(2\theta) = \frac{B}{A - C}.$$

The resulting equation can then be classified in the new coordinates using the standard forms above.

Degenerate Second-Degree Equations

Not every second-degree equation describes a non-degenerate conic. Factorisation or completing squares may reveal a simpler set.

Example

The equation

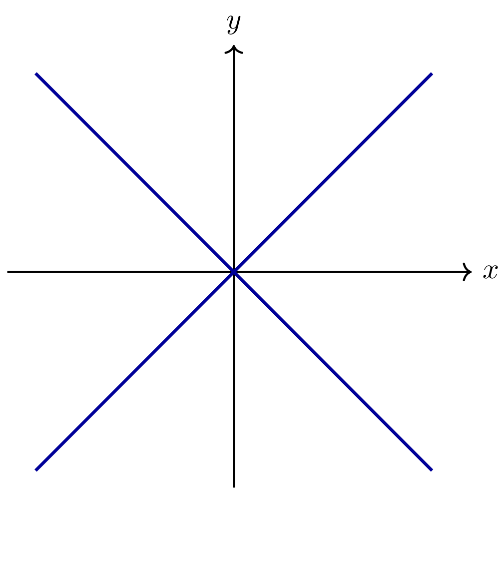
$$x^2 - y^2 = 0$$

factors as

$$(x - y)(x + y) = 0.$$

Its solution set is the union of two intersecting lines:

$$y = x \qquad \text{or} \qquad y = -x.$$



Other common degenerate cases include

$$x^2 + y^2 = 0,$$

which gives only the point $(0, 0)$,

$$x^2 + y^2 + 1 = 0,$$

which has no real solutions, and

$$x^2 = 0,$$

which describes the single line $x = 0$ counted algebraically twice.

Summary

Second-degree equations have the general form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

For non-degenerate real conics, the discriminant

$$\Delta = B^2 - 4AC$$

separates the three principal types:

$\Delta < 0$ gives ellipse type, $\Delta = 0$ gives parabola type, $\Delta > 0$ gives hyperbola type.

Translations and rotations reduce many equations to standard forms, while factorisation and completing squares expose degenerate cases.

4.2 Understanding Quadric Surfaces Through Sections

A three-dimensional surface should not be recognised only from a memorised formula. We study it by intersecting it with simple planes. Fixing $z = k$ gives a horizontal section; fixing $x = k$ or $y = k$ gives vertical sections. The remaining equation describes a planar curve already studied in this chapter.

The same questions will be asked repeatedly:

1. Which curve appears in a chosen cutting plane?
2. For which values of the parameter does that curve exist?
3. How does the section change as the plane moves?
4. What does that family of sections reveal about the whole surface?

This repetition is intentional. It is a method of investigation, not a list of unrelated recognition tricks.

4.2.1 Ellipsoid

For

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

the section $z = k$ satisfies

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 - \frac{k^2}{c^2}.$$

Thus $|k| < c$ gives an ellipse, $|k| = c$ gives one point, and $|k| > c$ gives no real points. The surface is understood as ellipses that grow from one endpoint, reach their largest size at $z = 0$, and shrink to the other endpoint.

4.2.2 Elliptic paraboloid

For
$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2},$$
the section $z = k$ is
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = k.$$
There are no points for $k < 0$, one vertex for $k = 0$, and growing ellipses for $k > 0$. Setting $y = 0$ gives the parabola
$$z = \frac{x^2}{a^2}.$$
The horizontal and vertical sections together explain both the opening direction and the shape near the vertex.

4.2.3 Hyperbolic paraboloid

For
$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2},$$
the horizontal section $z = k$ is
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = k.$$
For $k > 0$ the hyperbolas open in the x direction; for $k < 0$ they open in the y direction. At $k = 0$ the equation factors into two lines:
$$\left(\frac{x}{a} - \frac{y}{b}\right)\left(\frac{x}{a} + \frac{y}{b}\right) = 0.$$
The vertical sections $y = 0$ and $x = 0$ are parabolas opening in opposite directions. These changing sections explain the saddle shape.

4.2.4 Hyperboloid of one sheet

For
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1,$$
the section $z = k$ becomes
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 + \frac{k^2}{c^2}.$$
Every height gives an ellipse, smallest at $k = 0$ and larger as $|k|$ grows. Because a nonempty section exists at every height, the surface remains connected and has a central waist.

4.2.5 Hyperboloid of two sheets

For
$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$
the section $z = k$ satisfies
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{k^2}{c^2} - 1.$$
There are no points when $|k| < c$, one point when $|k| = c$, and ellipses when $|k| > c$. The empty band around $z = 0$ separates the surface into two components. The algebraic absence of sections explains the geometric gap.

4.2.6 Elliptic cone

For
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0,$$
the section $z = k$ is
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{k^2}{c^2}.$$
At $k = 0$ the section is one point. For positive and negative k we obtain ellipses of the same size whenever the absolute heights agree. This symmetry produces two nappes meeting at the origin.

4.2.7 Cylinders

If one coordinate is absent, moving in that coordinate direction never changes whether the equation is satisfied. For example,
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$
has the same elliptical section for every value of z. The ellipse is therefore carried unchanged along the z direction, producing an elliptic cylinder. Likewise,
$$y = x^2$$
produces the same parabola in every plane $z = k$, hence a parabolic cylinder.

Summary

A standard equation names a surface, but its sections explain the surface. Empty sections reveal gaps, shrinking sections reveal endpoints or vertices, changing conic types reveal saddle behaviour, and an absent coordinate reveals a direction of translation. The atlas below records the standard forms; this method explains why those forms have the geometry shown in their diagrams.

Atlas of Standard Quadric Surfaces

The three-dimensional analogue of a conic section is a *quadric surface*. Instead of an equation in two coordinates, we now allow a second-degree equation in three coordinates.

Definition

A second-degree equation in $\mathbb{R}^3$ has the form
$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0,$$
where at least one quadratic coefficient is nonzero. Its solution set is called a quadric surface when it forms a non-degenerate surface.

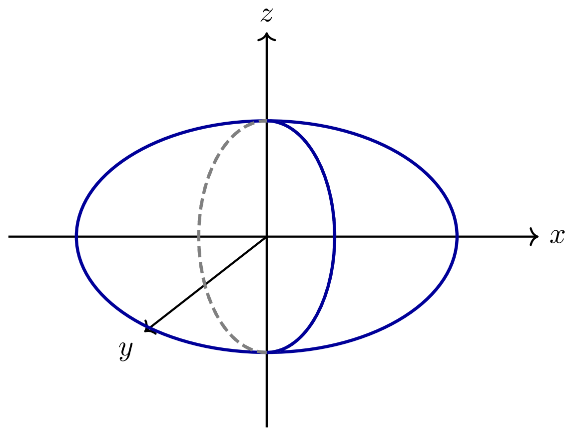
Translations move the centre or vertex of the surface. Rotations of the coordinate system remove mixed terms such as xy, xz, and yz. After these changes of coordinates, the principal quadric surfaces have simple standard forms.

Ellipsoid

An ellipsoid centred at the origin has equation
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \qquad a, b, c > 0.$$

Its intersections with the coordinate planes are ellipses. When $a = b = c = r$, the ellipsoid becomes the sphere

$$x^2 + y^2 + z^2 = r^2.$$

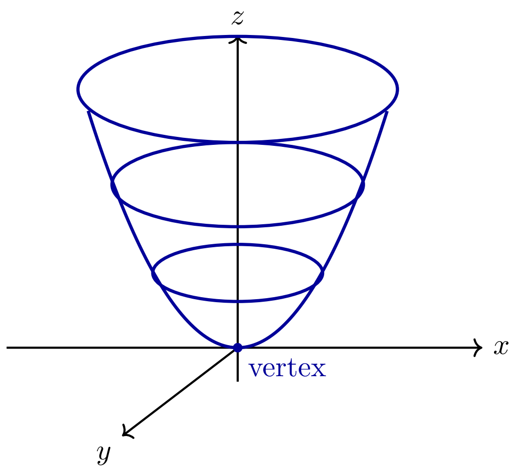


Elliptic Paraboloid

An elliptic paraboloid has standard equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}, \qquad a, b > 0.$$

Horizontal sections $z = k > 0$ are ellipses, while vertical sections through the z-axis are parabolas. The surface has one vertex and opens in one direction.

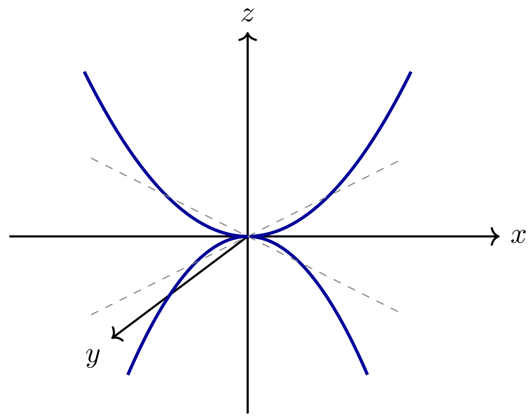


Hyperbolic Paraboloid

A hyperbolic paraboloid has standard equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

Sections parallel to the xz-plane and yz-plane are parabolas opening in opposite directions. Horizontal sections are hyperbolas, except at $z = 0$, where the section consists of two intersecting lines. The surface is commonly called a saddle.

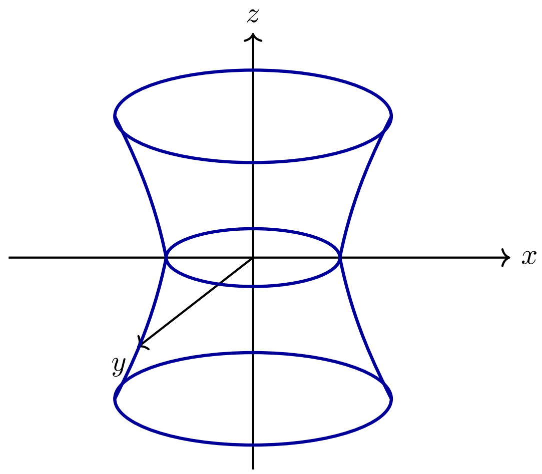


Hyperboloid of One Sheet

A hyperboloid of one sheet has standard equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1.$$

Horizontal sections are ellipses. Vertical sections through the z-axis are hyperbolas. The surface is connected and has a narrow central waist.

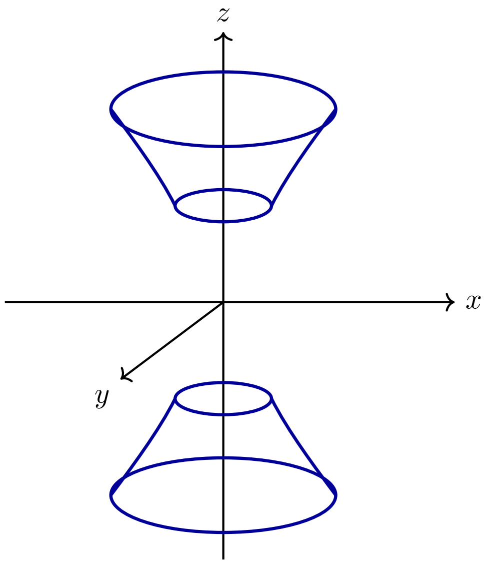


Hyperboloid of Two Sheets

A hyperboloid of two sheets has standard equation

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

The condition forces $|z| \geq c$, so the surface has two disconnected parts. Horizontal sections are ellipses, while suitable vertical sections are hyperbolas.

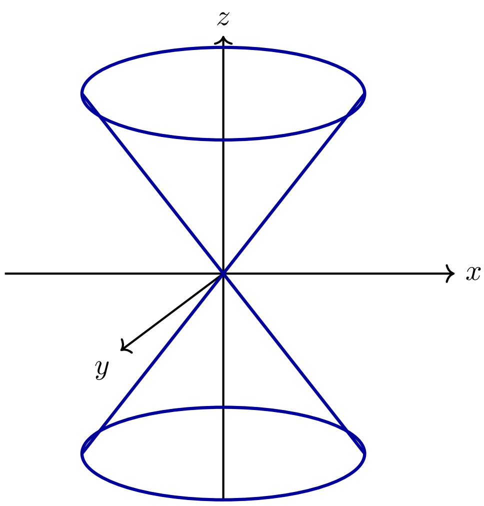


Elliptic Cone

An elliptic cone has equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

It consists of two nappes meeting at the origin. Horizontal sections away from the origin are ellipses. Vertical sections through the axis are pairs of intersecting lines.



Cylinders as Second-Degree Surfaces

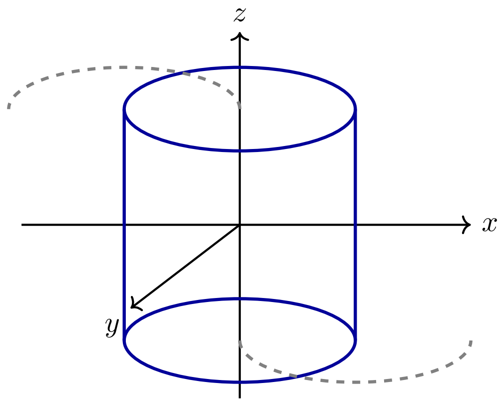
If one coordinate does not occur in the equation, the corresponding planar curve is extended parallel to that coordinate axis. For example,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

describes an elliptic cylinder parallel to the z-axis, while

$$y = x^2$$

describes a parabolic cylinder parallel to the z-axis.



Degenerate Quadrics

As in the planar case, a second-degree equation in three variables may be degenerate. It may describe a point, a line, a plane, two planes, or no real points. For example,

$$x^2 + y^2 + z^2 = 0$$

describes only the origin, while

$$x^2 - y^2 = 0$$

factors as

$$(x - y)(x + y) = 0$$

and therefore describes the union of the two planes

$$x - y = 0 \qquad \text{and} \qquad x + y = 0.$$

Summary

Second-degree equations in three variables produce quadric surfaces. After translations and rotations, the main non-degenerate types are ellipsoids, elliptic and hyperbolic paraboloids, hyperboloids of one or two sheets, and elliptic cones. Equations missing one coordinate give cylinders, while factorisation may reveal degenerate unions of planes or smaller sets.

Chapter 5

Chapter 6

Chapter 7

Chapter 8

Chapter 9

Chapter 10

Chapter 11

Chapter 12